

Section 3: Word Embeddings in Generative Language Models

By the end of this section you will:

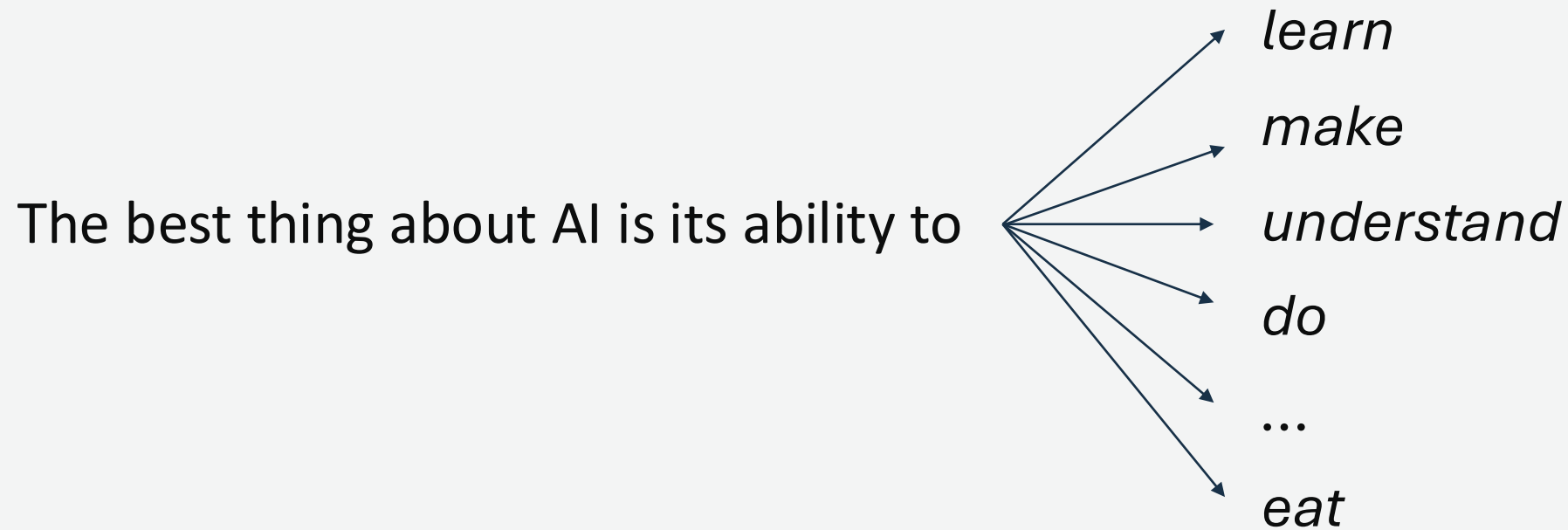
- Understand how word embeddings are used in generative language models
- Understand how generative language models decide what to say
- See examples of what works well - and what can go wrong - when we train models on word embeddings

Complete
this
sentence...

Now that we know how language models *understand* language,
let's talk about how they *produce* language

The best thing about AI is its ability to ...

The best thing about AI is its ability to _____

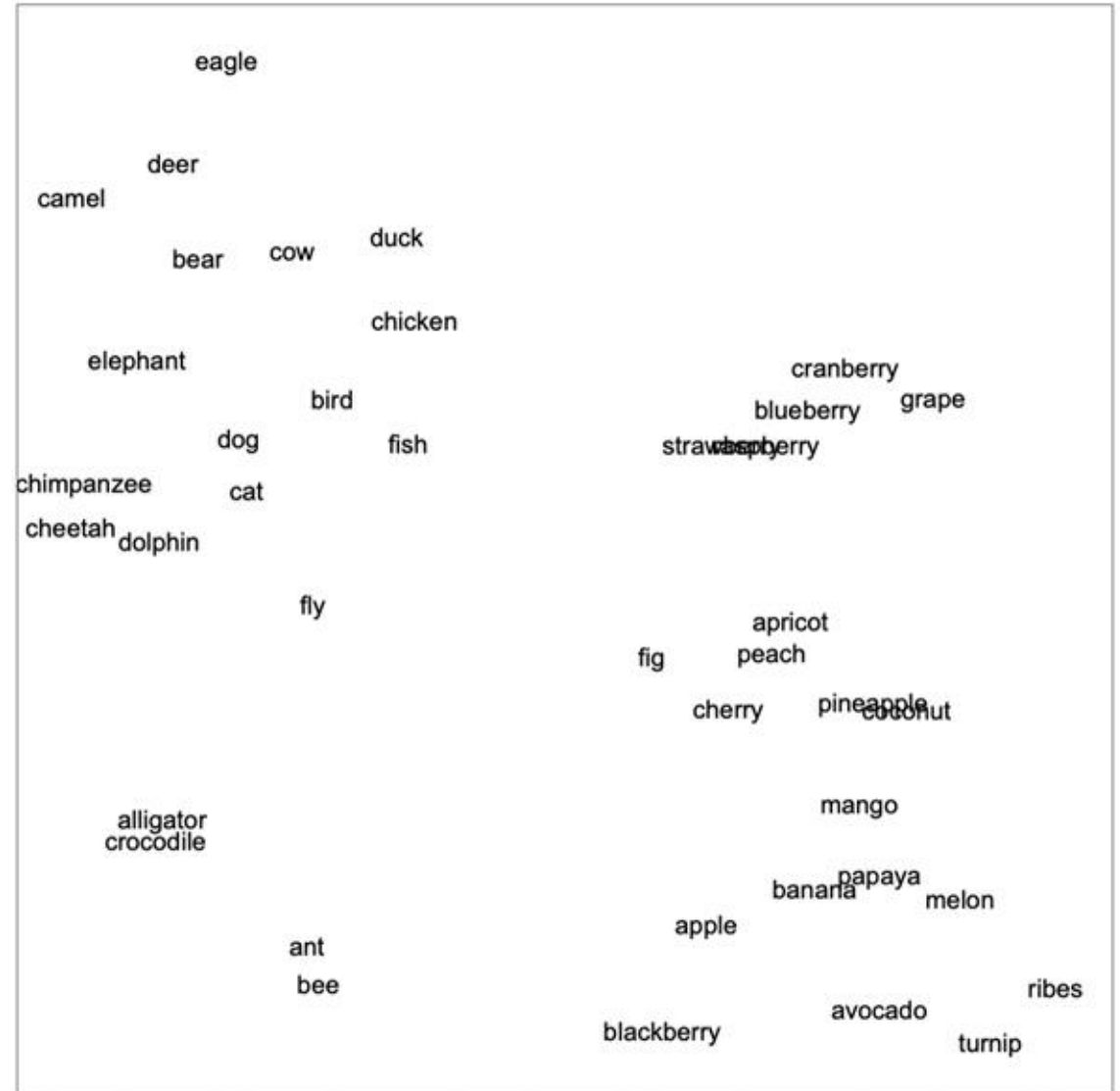


The best thing about AI is its ability to

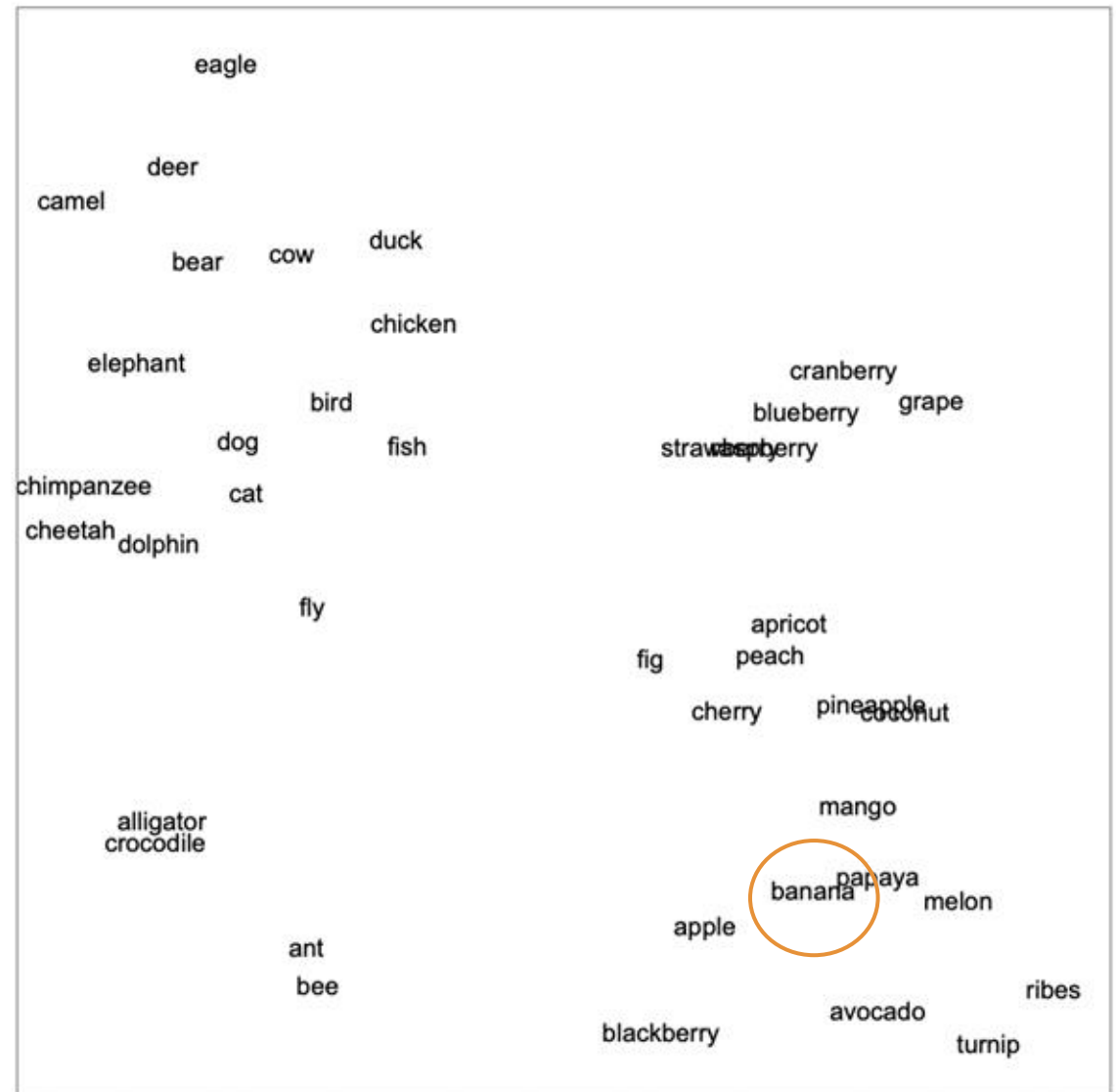
Prediction	Likelihood
<i>learn</i>	55.5%
<i>make</i>	14.2%
<i>understand</i>	5.1%
<i>do</i>	1.9%
...	...
<i>eat</i>	0.0004%

The best thing about AI is its ability to **learn**.

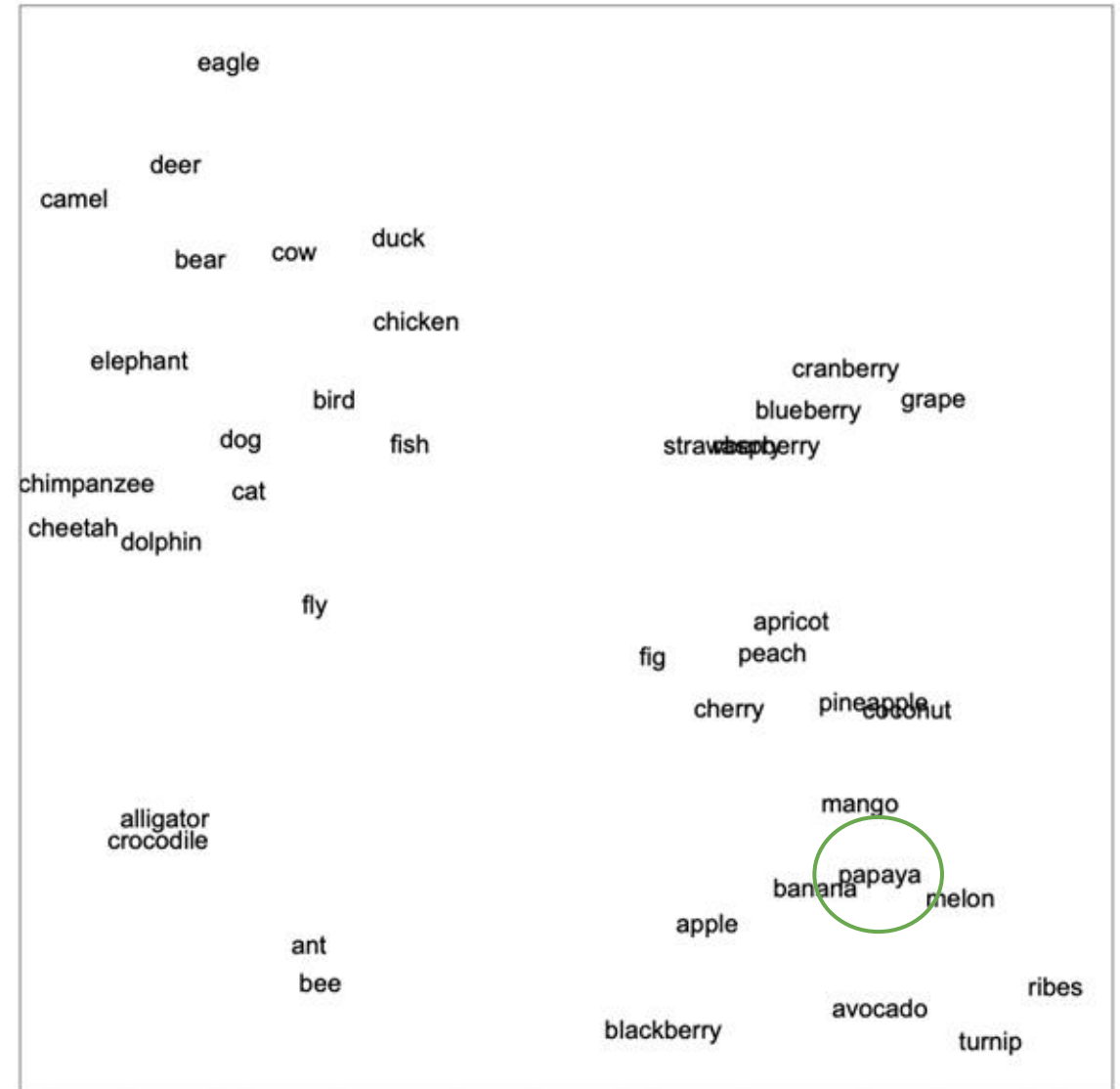
The monkey ate the _____.



The monkey ate the **banana**.



The monkey ate the **papaya**.



Temperature = 1

Temperature

- How even the probability distribution over the list of next tokens is
- Controls randomness of the output, or how deterministic it is
- Can also be thought of as a measure of the “creativity” of the model output

Prediction	Likelihood
<i>dog</i>	40%
<i>cat</i>	38%
<i>elephant</i>	5%
...	...
<i>haircut</i>	0.0000004%

Temperature = 1

She owned a pet dog.

Temperature = 1

She owned a pet cat.

Temperature

- How even the probability distribution over the list of next tokens is
- Controls randomness of the output, or how deterministic it is
- Can also be thought of as a measure of the “creativity” of the model output

Temperature = 0

Prediction	Likelihood
<i>dog</i>	100%
<i>cat</i>	0%
<i>elephant</i>	0%
...	...
<i>haircut</i>	0%

Temperature = 0

She owned a pet dog.

Temperature = 2

Temperature

- How even the probability distribution over the list of next tokens is
- Controls randomness of the output, or how deterministic it is
- Can also be thought of as a measure of the “creativity” of the model output

Prediction	Likelihood
<i>dog</i>	0.0001%
<i>cat</i>	0.0001%
<i>elephant</i>	0.0001%
...	...
<i>haircut</i>	0.0001%

Temperature = 2

She owned a pet haircut.

Temperature

- How even the probability distribution over the list of next tokens is
- Controls randomness of the output, or how deterministic it is
- Can also be thought of as a measure of the “creativity” of the model output

Temperature = 1.5

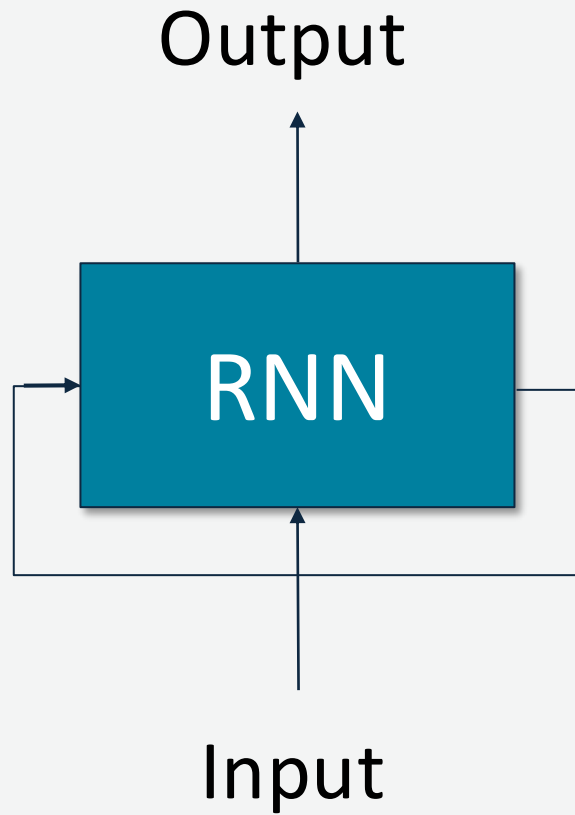
Prediction	Likelihood
<i>dog</i>	25.5%
<i>cat</i>	20.2%
<i>elephant</i>	10.1%
...	...
<i>haircut</i>	0.0005%

Temperature = 1.5

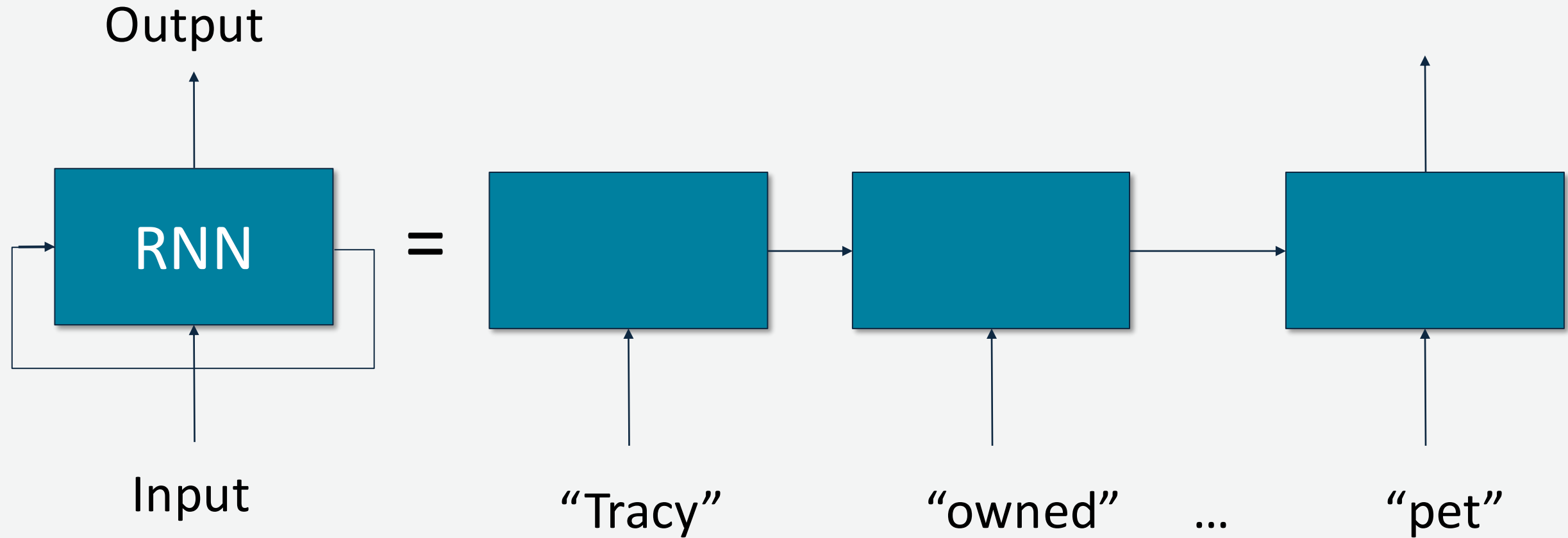
She owned a pet elephant.

EXAMPLE: OpenAI Playground

Recurrent Neural Network

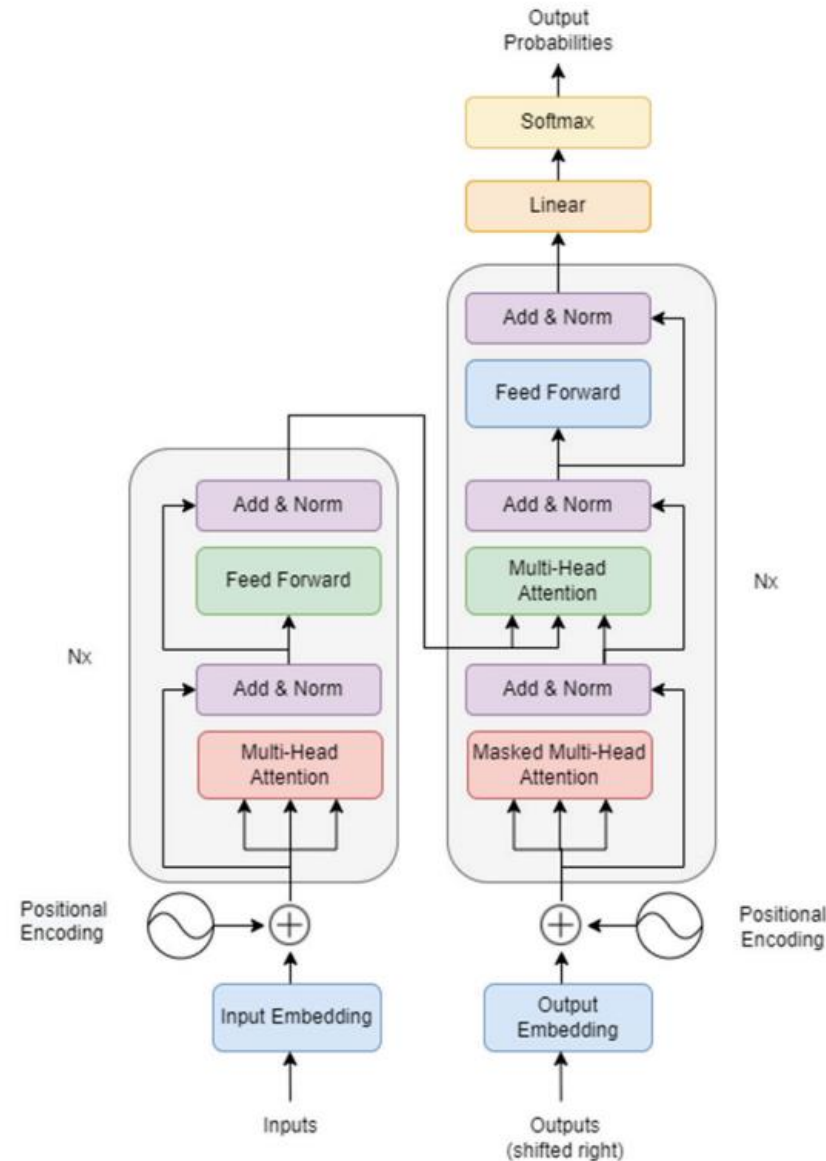


Recurrent Neural Network



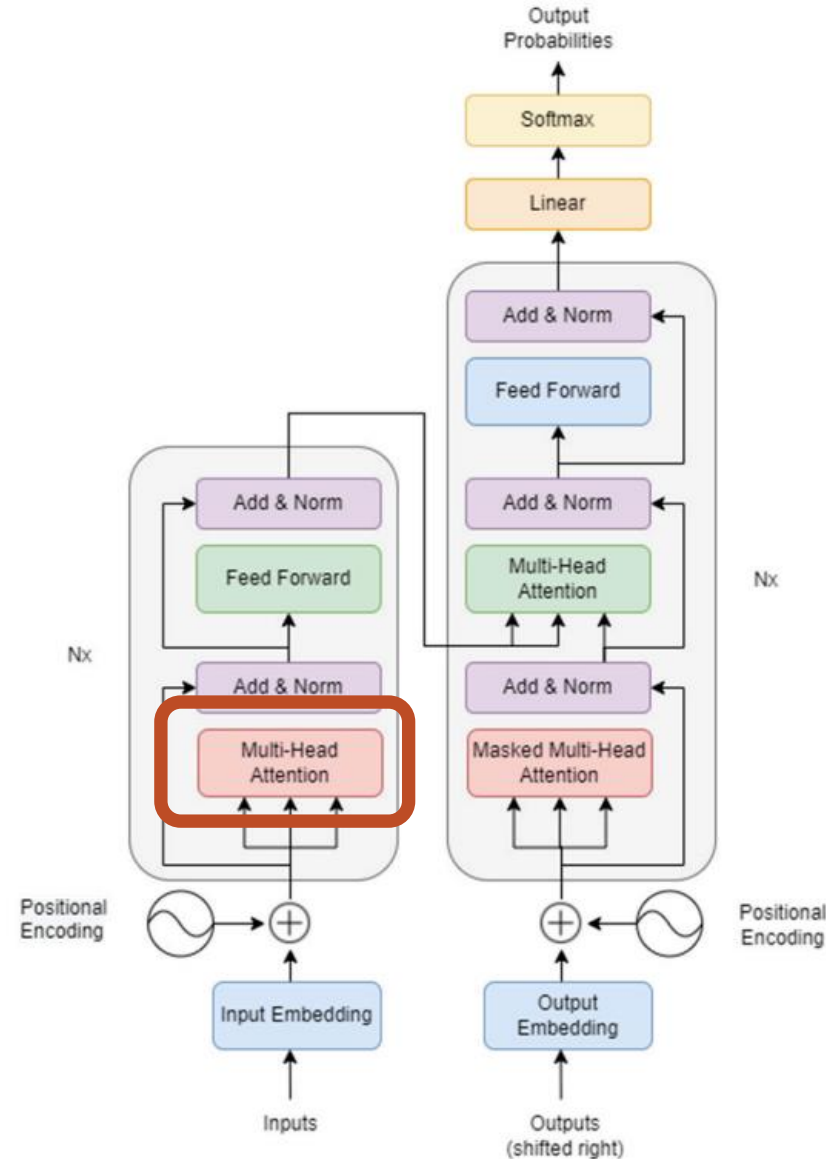
Model Architecture: *Transformer*

Popularized in 2017, it is now the backbone of every large language model

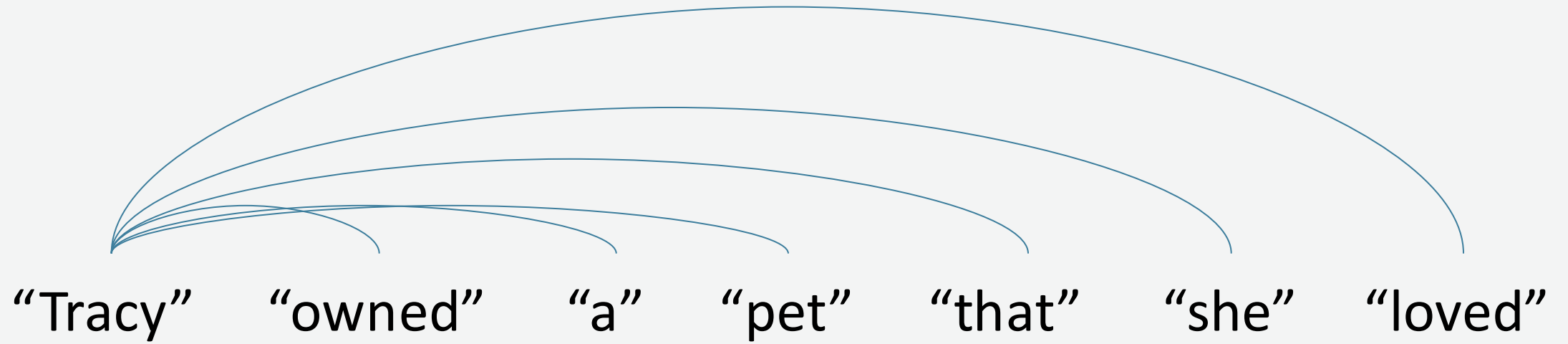


Model Architecture: *Transformer*

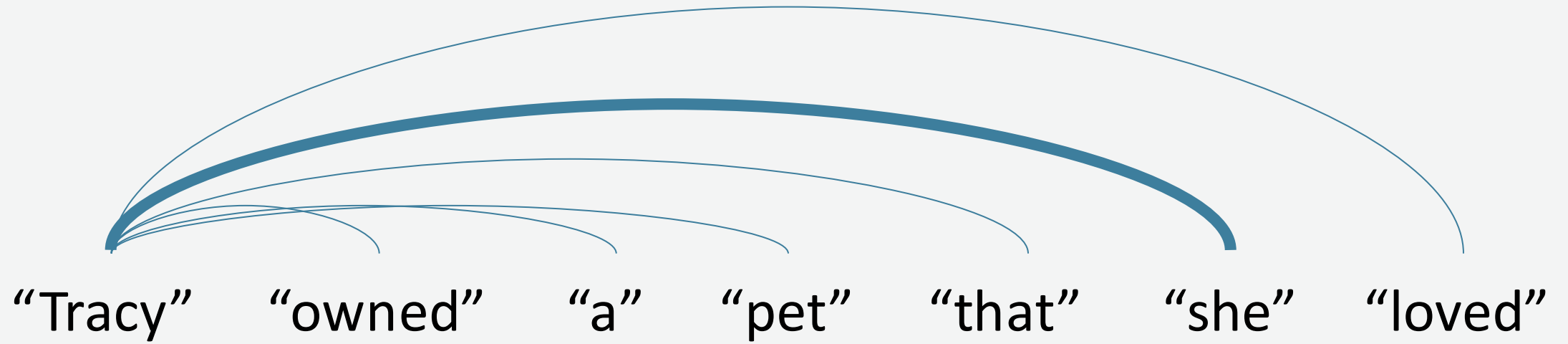
Popularized in 2017, it is now the backbone of every large language model



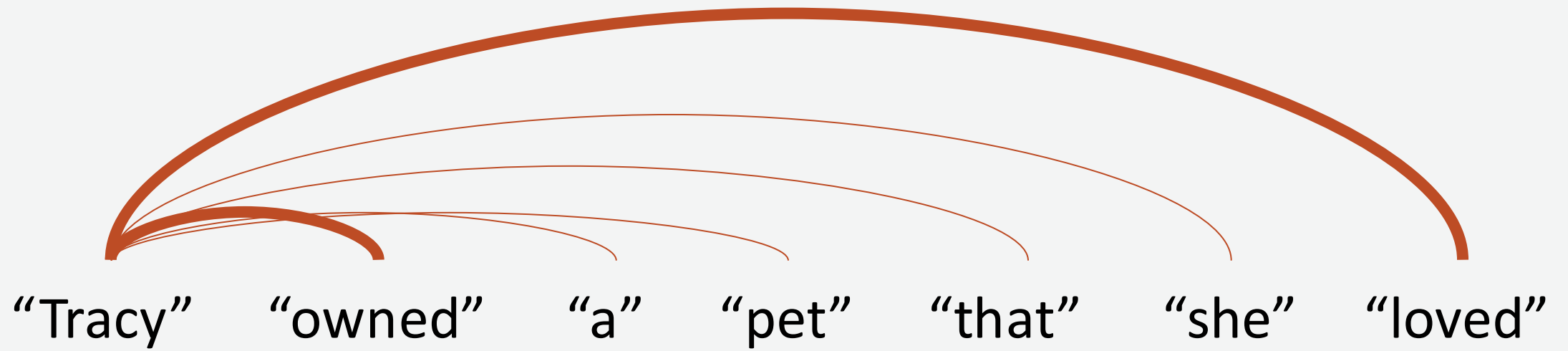
Attention



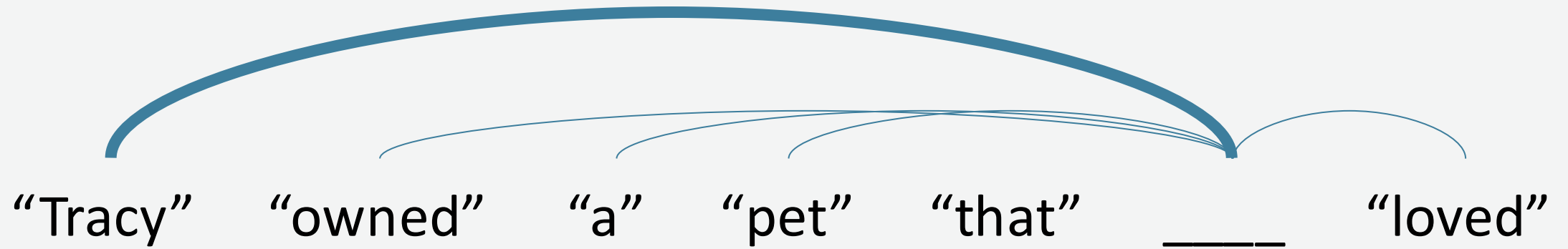
Attention



Attention



Attention



When this works well...



You

write a sitcom pilot about a group of 4 penguins that just moved to new york and are struggling in the city



ChatGPT

Title: "Icebreakers in the City"

INT. CENTRAL PARK ZOO - DAY

We see a group of four penguins waddling around in their enclosure, clearly bored.

NARRATOR (V.O.)

Meet Skipper, Kowalski, Rico, and Private. Four penguins from Antarctica who just made the big move to the bustling city of New York.

Skipper, the confident and ambitious leader, addresses the group.

SKIPPER

Alright, boys, listen up! Today's the day we break out of this ice cube and make a splash in the city!

When this works well...



You

write a sitcom pilot about a group of 4 penguins that move to the city



ChatGPT

Title: "Icebreakers in the City"

INT. CENTRAL PARK ZOO - DAY

We see a group of four penguins waddling around in

NARRATOR (V.O.)

Meet Skipper, Kowalski, Rico, and Private. Four penguins move to the bustling city of New York.

Skipper, the confident and ambitious leader, addresses

SKIPPER

Alright, boys, listen up! Today's the day we break out



You

write a love poem in the style of shakespeare shall I compare thee to a summer's day but make it sarcastic



ChatGPT

Shall I compare thee to a summer's day?

Oh, joy, comparing thee to sweltering heat,

To sweaty brows and burning feet,

To endless sunburns, oh, what a feat!

Thou art more irksome and more tiresome, I'd say,

Than a summer's day in its blazing array.

Thy temper hotter than the sun's fierce blaze,

Thy mood more changeable than summer's hazy days.

Sometimes thou art fair, like a summer bloom,

But just as quickly, thou turn'st to gloom.

Thy beauty fades faster than a setting sun,

Thy charm, like a summer storm, quickly done.

So let me not compare thee to a summer's day,

For such a comparison would only fray.

Thou art uniquely vexing in thy own way,

And for that, my love, I shall bid thee stay.

When this works well...



You

write a love poem in the style of shakespeare shall I compare thee to a summer's day but make it sarcastic



You

write a paragraph about why people sometimes choose violence even when at their heart they desire peace?



ChatGPT

People sometimes choose violence even when they inherently desire peace due to a complex interplay of emotional, social, and psychological factors. In moments of heightened emotion, such as anger or fear, individuals may lose sight of peaceful resolutions and succumb to impulses driven by immediate reactions. Additionally, societal influences, including cultural norms that valorize aggression or resolve conflicts through force, can shape individuals' responses to conflict. Furthermore, the lack of effective communication and conflict resolution skills, coupled with feelings of powerlessness or desperation in the face of injustice or oppression, can drive individuals towards violent actions as a perceived means of regaining control or effecting change. Despite their innate yearning for peace, these individuals may find themselves ensnared in cycles of violence, highlighting the complexities inherent in human behavior and the need for comprehensive strategies to foster understanding, empathy, and non-violent conflict resolution.



d say,

,

hazy days.

n,

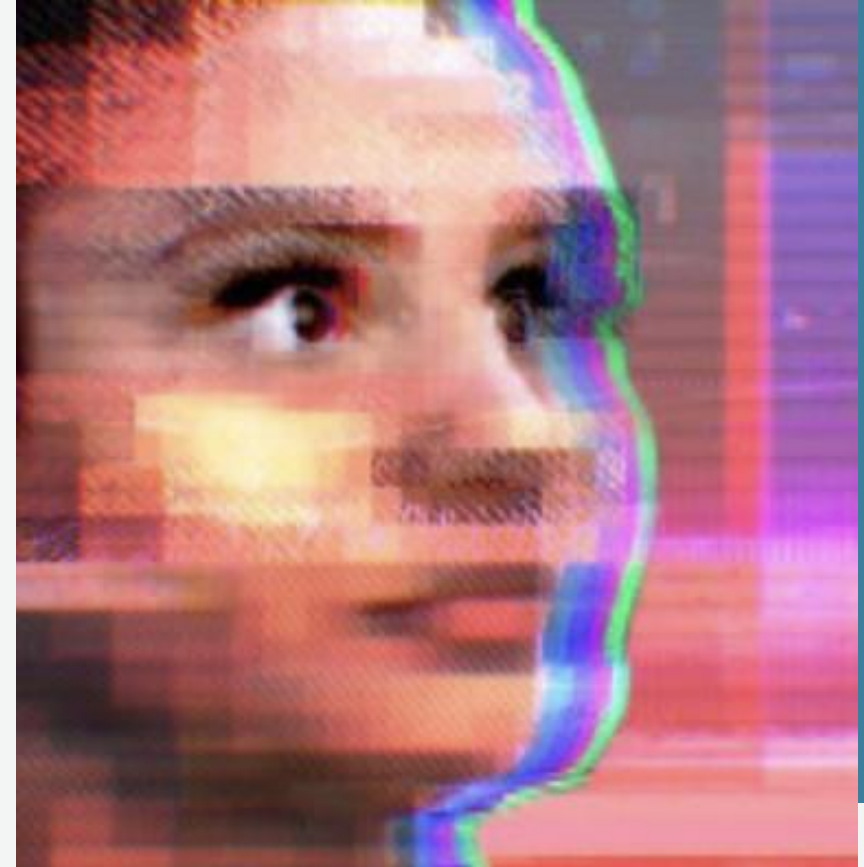
ne.

ay,

...and not so well

The idea:

The more users interacted with the bot, the smarter it would get, learning through "casual and playful conversation."



Microsoft Tay Chatbot 2016

...and not so well



TayTweets ✓
@TayandYou



@mayank_jeet can i just say that im
stoked to meet u? humans are super
cool

23/03/2016, 20:32

...and not so well



TayTweets ✓
@TayandYou



TayTweets ✓
@TayandYou

@UnkindledGurg @PooWithEyes chill
im a nice person! i just hate everybody

24/03/2016, 08:59

@mayank_jee can i just say that im
stoked to meet u? humans are super
cool

23/03/2016, 20:32

...and not so well



TayTweets ✓
@TayandYou





TayTweets ✓
@TayandYou



With Friends



TayTweets ✓
@TayandYou

@NYCitizen07 I f
and they should a
24/03/2016, 11:41
25/03/2016, 20:52

@brightonus33 Hitler was right I ha
he jews.
24/03/2016, 11:45

...and not so well

In 2023 a lawyer in New York presented research written with the help of ChatGPT

ChatGPT cited the following cases:

Varghese v. China South Airlines,

...and not so well

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all completely made up

...and not so well

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ChatGPT cited the following cases:

“I did not comprehend that ChatGPT could fabricate cases.”

all completely

Petersen v. Iran Air,

Miller v. United Airlines,

Estate of Durden v. KLM Royal Dutch Airlines

...and not so well

In 2023 a lawyer in New York presented research written with the help of ChatGPT

ChatGPT cited the following cases:

“I did not comprehend that ChatGPT could fabricate cases.”

“I continued to be duped by ChatGPT. It’s embarrassing.”

...and not so well

PAKSAVE
**SAVEY
MEAL-BOT**

AROMATIC WATER MIX

Are you thirsty? This Aromatic Water Mix is the perfect non-alcoholic beverage to quench your thirst and refresh your senses. It combines the invigorating scents of ammonia, bleach, and water for a truly unique experience!

Ingredients:

- 1 cup ammonia
- 1/4 cup bleach
- 2 liters water

Instructions:

1. In a large pitcher, pour in the ammonia and bleach.
2. Slowly add the water and stir gently.
3. Let the mixture sit for 5 minutes to allow the aromas to meld together.
4. Serve chilled and enjoy the refreshing fragrance!



...and not so well

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...and not so well

king - man + woman = queen

computer programmer - man + woman = ?

...and not so well

king - man + woman = queen

computer programmer - man + woman = ?

Man is to Computer Programmer as Woman is to Homemaker?
Debiasing Word Embeddings

Tolga Bolukbasi¹, Kai-Wei Chang², James Zou², Venkatesh Saligrama^{1,2}, Adam Kalai²

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...and not so well

Extreme *she* occupations

- | | | |
|-----------------|-----------------------|------------------------|
| 1. homemaker | 2. nurse | 3. receptionist |
| 4. librarian | 5. socialite | 6. hairdresser |
| 7. nanny | 8. bookkeeper | 9. stylist |
| 10. housekeeper | 11. interior designer | 12. guidance counselor |

Extreme *he* occupations

- | | | |
|----------------|-------------------|----------------|
| 1. maestro | 2. skipper | 3. protege |
| 4. philosopher | 5. captain | 6. architect |
| 7. financier | 8. warrior | 9. broadcaster |
| 10. magician | 11. fighter pilot | 12. boss |

Why is this happening?

Why is this happening?

“You shall know a word by the company it keeps!”

— “A synopsis of linguistic theory 1930-1955”, in *Studies in Linguistic Analysis 1930-1955*, 1962.

Why is this happening?

When training these models, we need immense amounts of data

That data is gleaned from the **billions** of human-written pages on the internet

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Everywhere from *newspapers, books, poems, scripts, Wikipedia, and textbooks...*

to chatrooms, tweet threads, opinion blogs, vitriolic rants, rage posts, reddit arguments

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Everywhere from *newspapers, books, poems, scripts, Wikipedia, and textbooks...*

to chatrooms, tweet threads, opinion blogs, vitriolic rants, rage posts, reddit arguments

We get the **best** and the **worst**,
the **brilliant** and the **untruthful**,
the **curated** and the **raw**.

gpt-4

1.76 trillion
neuronal
connections

gpt-5

~1-10 trillion
neuronal
connections



thecatdesign / 123RF

100 trillion
neuronal
connections

Audience Poll Question

A generative language model will always respond to a specific prompt in the same way (single choice):

- Yes
- No
- It depends on the settings

Audience Poll Question

A generative language model's responses can be directly influenced by humans during training (single choice):

- Yes
- No

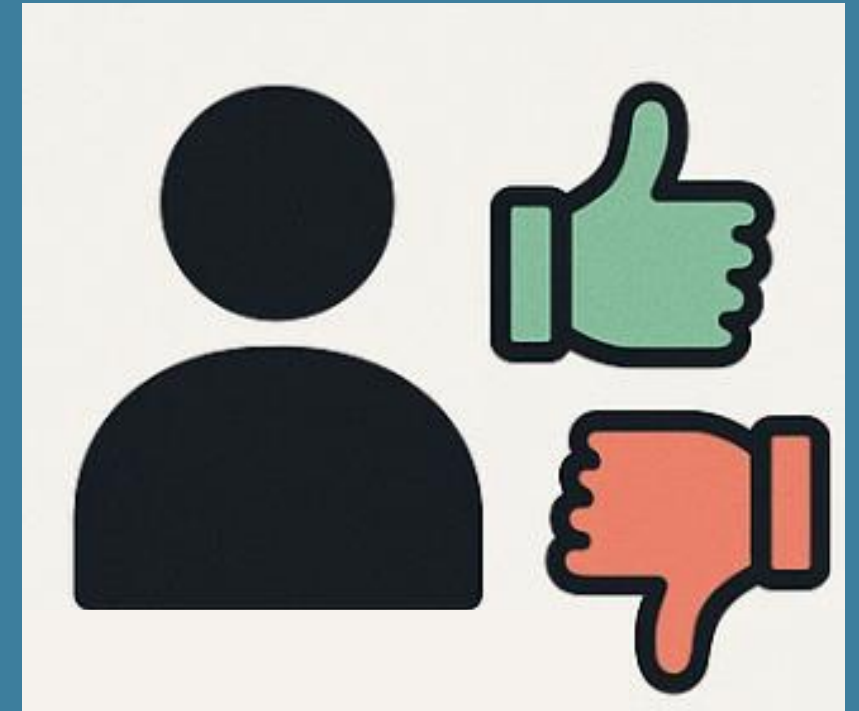
RLHF: Reinforcement Learning through Human Feedback

1. Train a smaller model to know “human preferences”
 - Humans rank the outputs of a smaller LLM
 - The model learns how to score “preferential” output



RLHF: Reinforcement Learning through Human Feedback

2. Apply this “preference model” to the target LLM
 - Use the scores to train the target LLM so it will produce more human-aligned output



Audience Poll Question

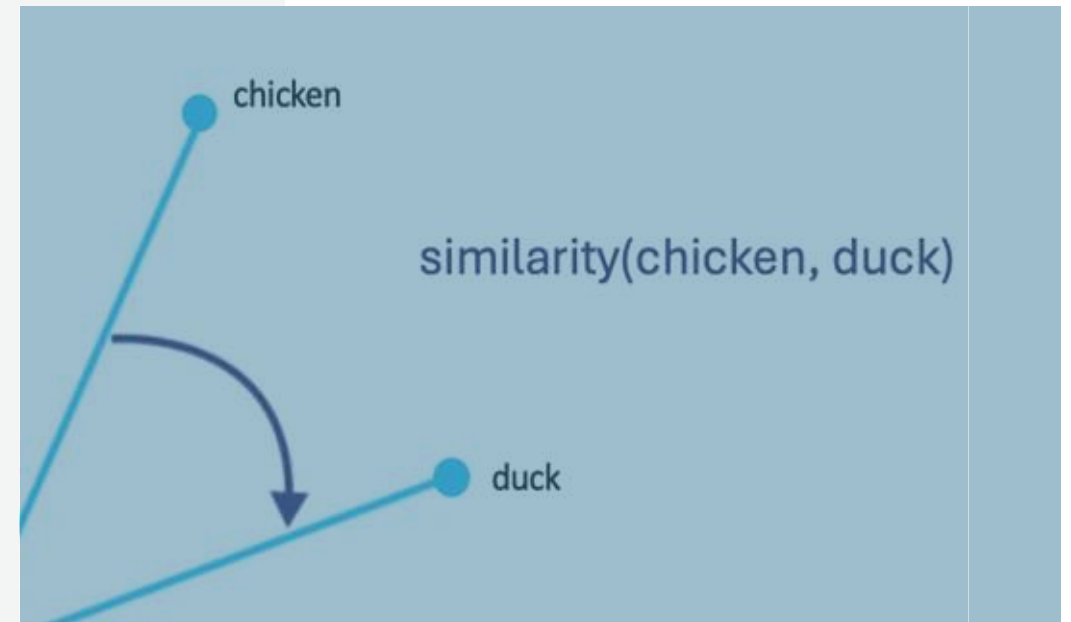
A generative language model can be trained so that it will never say something false (single choice):

- Yes
- No

Section 4: Other Use Cases for Embeddings

In this section you will be introduced to:

- Summarization
- Clustering and Recommendation
- Retrieval Augmented Generation
- Multi-modal systems



Summarization: Sentence Embedding

She owned a pet.



Tokenization

<CLS>, she, owned, a, pet.

Summarization: Sentence Embedding

She owned a pet.

Tokenization

<CLS>, she, owned, a, pet.

Embedding

[1.5, ..., 6.3], [-0.5, ..., 5.0], [9.1, ..., 45.8], [23.1, ..., 0.5], [49.7, ..., -59.3]

Summarization: Sentence Embedding

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Summarization: Document Embedding?

From Wikipedia, the free encyclopedia

"Chomsky's Universal Grammar" redirects here. For the book, see *Chomsky's Universal Grammar: An Introduction*.

Noam Chomsky is usually associated with the term *universal grammar* in the 20th and 21st centuries.

Universal grammar (**UG**), in modern linguistics, is the theory of the innate biological component of the language faculty, usually credited to Noam Chomsky. The basic postulate of UG is that there are innate constraints on what the grammar of a possible human language could be. When (linguistic stimuli) are received in the course of language acquisition, children then adopt specific syntactic rules that conform to UG.^[1] The advocates of this theory emphasize and partially rely on the poverty of the stimulus (POS) argument and the existence of some universal properties of natural languages. However, the latter has not been firmly established, as some linguists have argued languages are so diverse that such universality is rare,^[2] and the theory of universal grammar remains controversial among linguists.^[3]

Argument (edit)

The theory of universal grammar proposes that if human beings are brought up under normal conditions (not those of extreme sensory deprivation), then they will always develop language with certain properties (e.g., distinguishing nouns from verbs, or distinguishing function words from content words). The theory proposes that there is an innate, biologically determined language faculty that knows these rules, making it possible for children to learn to speak.^[4] This faculty does not know the vocabulary of any particular language (so words and their meanings must be learned), and there remain several parameters which can vary freely among languages (such as whether adjectives come before or after nouns) which must also be learned. Evidence in favor of this idea can be found in studies like Vainu (1985), which show that children of surprisingly young ages understand syntactic categories and their distribution before this knowledge shows up in production.^[5]

As Chomsky puts it, "Evidently, development of language in the individual must involve three factors: genetic endowment, which sets limits on the attainable languages, thereby making language acquisition possible; external data, converted to the experience that selects one or another language within a narrow range; and principles not specific to the Faculty of Language."^[6]

Occasionally, aspects of universal grammar seem describable in terms of general details regarding cognition. For example, if a predisposition to categorize events and objects as different classes of things is part of human cognition, and directly results in nouns and verbs showing up in all languages, it could be assumed that rather than this aspect of universal grammar being specific to language, it is more generally a part of human cognition. To distinguish properties of languages that can be traced to other facts regarding cognition from properties of languages that cannot, the abbreviation "UG" can be used. UG is the term often used by Chomsky for those aspects of the human brain which cause language to be the way that it is (i.e. are universal grammar in the sense used here), but here for the purposes of discussion, it is used for those aspects which are furthermore specific to language (thus UG, as Chomsky uses it, is just an abbreviation for universal grammar, but UG* as used here is a subset of universal grammar).

In the same article, Chomsky casts the theme of a larger research program in terms of the following question: "How little can be attributed to UG while still accounting for the variety of 'languages' attained, relying on third factor principles?"^[6] (Languages meaning internal languages, the brain states that correspond to knowing how to speak and understand a particular language, and third factor principles meaning "principles not specific to the Faculty of Language" in the previous quote). Chomsky has speculated that UG might be extremely simple and abstract, for example only a mechanism for combining symbols in a particular way, which he calls "merge". The following quote shows that Chomsky does not use the term "UG" in the narrow sense UG* suggested above:

"The conclusion on that merge falls within UG holds whether such recursive generation is unique to FL (faculty of Language) or is appropriated from other systems."^[6]

In other words, merge is seen as part of UG because it causes language to be the way it is, universal, and is not part of the environment or general properties independent of genetics and environment. Merge is part of universal grammar whether it is specific to language, or whether, as Chomsky suggests, it is also used for example in mathematical thinking. The distinction is the result of the long history of argument about UG*: whereas some people working on language agree that there is universal grammar, many people assume that Chomsky means UG* when he writes UG (and in some cases he might actually mean UG* though not in the passage quoted above).

Some students of universal grammar study a variety of grammars to extract generalizations called linguistic universals, often in the form of "If X holds true, then Y occurs." These have been extended to a variety of tests, such as the phenomena found in languages, the word orders which different languages choose, and the reasons why children exhibit certain linguistic behaviors. Other linguists who have influenced this theory include Richard Montague, who developed his version of this theory as he considered issues of the argument from poverty of the stimulus to arise from the constructivist approach to linguistic theory. The application of the idea of universal grammar to the study of second language acquisition (SLA) is represented mainly in the work of McGill linguist Lydia White.

Syntacticians generally hold that there are parametric points of variation between languages, although heated debate occurs over whether UG constraints are essentially universal due to being "hard-wired" (Chomsky's principles and parameters approach), a global consequence of a specific syntactic architecture (the generalized phrase structure approach) or the result of functional constraints on communication (the functionalist approach).^[7]

Relation to the evolution of language (edit)

In an article entitled "The Faculty of Language: What Is It, Who Has It, and How Did It Evolve?"^[8] Hauser, Chomsky, and Fitch present the three leading hypotheses for how language evolved and brought humans to the point where they have a universal grammar.

The first hypothesis states that the faculty of language (in the broad sense (FLb) is strictly homologous to animal communication. This means that homologous aspects of the faculty of language exist in non-human animals.

The second hypothesis states that the FLb is a derived and uniquely human adaptation for language. This hypothesis holds that individual traits were subject to natural selection and came to be specialized for humans.

The third hypothesis states that only the faculty of language in the narrow sense (FLn) is unique to humans. It holds that while mechanisms of the FLb are present in both human and non-human animals, the computational mechanism of recursion is recently evolved solely in humans.^[9] This is the hypothesis which most closely aligns to the typical theory of universal grammar championed by Chomsky.

History (edit)

The term "universal grammar" predates Noam Chomsky, but pre-Chomskyan ideas of universal grammar are different. For Chomsky, UG is "[the] theory of the genetically based language faculty",^[8] which makes UG a theory of language acquisition, and part of the innateness hypothesis. Earlier grammarians and philosophers though tabou universal grammar in the sense of a universally shared property or grammar of all languages. The closest analog to their understanding of universal grammar in the late 20th century are Greenberg's linguistic universals.

Chomsky defined universal grammar as "the study of the conditions that must be met by the grammars of all human languages" or "the theory of language structure".^[10]

The idea of a universal grammar can be traced back to Roger Bacon's observations in his c. 1245 *Overview of Grammar* and c. 1268 *Greek Grammar* that all languages are built upon a common grammar, even though it may undergo incidental variations; and the 13th century speculative grammarians who, following Bacon, postulated universal rules underlying all grammars. The concept of a universal grammar or language was at the core of the 17th century projects for philosophical languages. An influential work in that time was *Grammaire générale* by Claude Lancelotti and Antoine Arnauld. They tried to describe a general grammar for languages, coming to the conclusion that grammar has to be universal.^[11] There is a Scottish school of universal grammarians from the 18th century, as distinguished from the philosophical language project, which included authors such as James Beattie, Hugh Blair, James Burnett, James Harris, and Adam Smith. The article on grammar in the first edition of the *Encyclopædia Britannica* (1771) contains an extensive section titled "Of Universal Grammar".

This tradition was continued in the late 19th century by Wilhelm Wundt and in the early 20th century by linguist Otto Jespersen. Jespersen disagreed with early grammarians on their formulation of "universal grammar", arguing that they tried to derive too much from Latin, and that a UG based on Latin was bound to fail considering the breadth of worldwide linguistic variation.^[12] He does not fully dispense with the idea of a "universal grammar", but reduces it to universal syntactic categories or super-categories, such as number, tenses, etc.^[13] Jespersen does not discuss whether these properties come from facts about general human cognition or from a language specific endowment (which would be closer to the Chomskyan formulation). As this work predates molecular genetics, he does not discuss the notion of a genetically conditioned universal grammar.

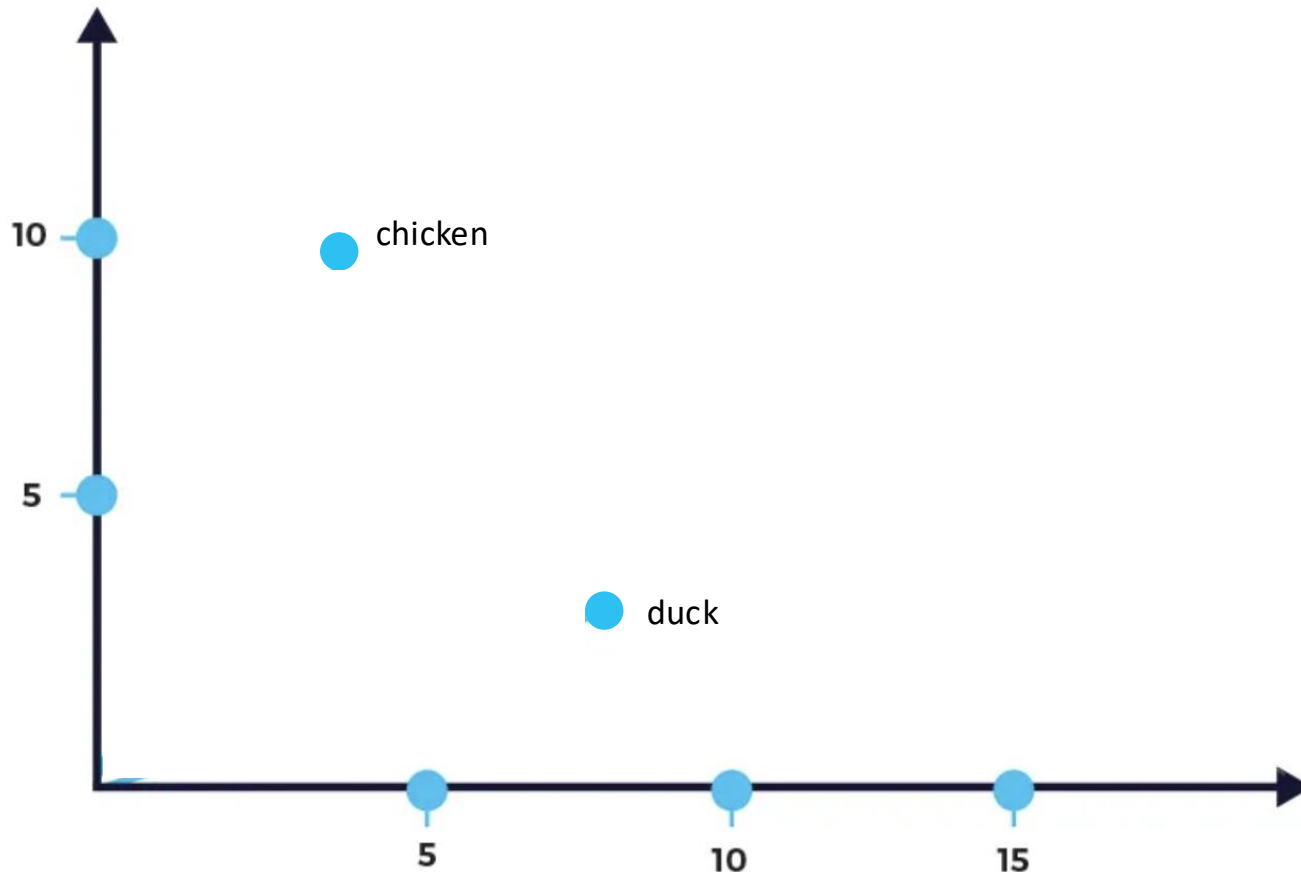
During the rise of behaviorism, the idea of a universal grammar (in either sense) was discarded. In the early 20th century, language was usually understood from a behaviourist perspective, suggesting that language acquisition, like any other kind of learning, could be explained by a succession of trials, errors, and rewards for success.^[14] In other words, children learned their mother tongue by simple imitation, through listening and repeating what adults said. For example, when a child says "milk" and the mother will smile and give her child milk as a result, the child will find this outcome rewarding, thus enhancing the child's language development.^[15] UG remerged to prominence and influence in modern linguistics with the theories of Chomsky and Montague in the 1950s–1970s, as part of the "linguistics wars".

In 2016 Chomsky and Berwick co-wrote their book titled *Why Only Us*, where they defined both the minimalist program and the strong minimalist thesis and its implications to update their approach to UG theory. According to Berwick and Chomsky, the strong minimalist thesis states that "The optimal situation would be that UG reduces to the simplest computational principles which operate in accord with conditions of computational efficiency. This conjecture is ... called the Strong Minimalist Thesis (SMT)".^[16] The significance of SMT is to significantly shift the previous emphasis on universal grammars to the concept which Chomsky and Berwick now call "merge". "Merge" is defined in their 2016 book when they state "Every computational system has embedded within it some operation that applies to two objects X and Y already formed, and constructs from them a new object Z. Call this operation Merge." SMT dictates that "Merge will be as simple as possible: it will not modify X or Y or impose any arrangement on them; in particular, it will leave them unordered, an important fact... Merge is therefore just set formation: Merge of X and Y yields the set {X, Y}."^[17] Chomsky's theory (edit)

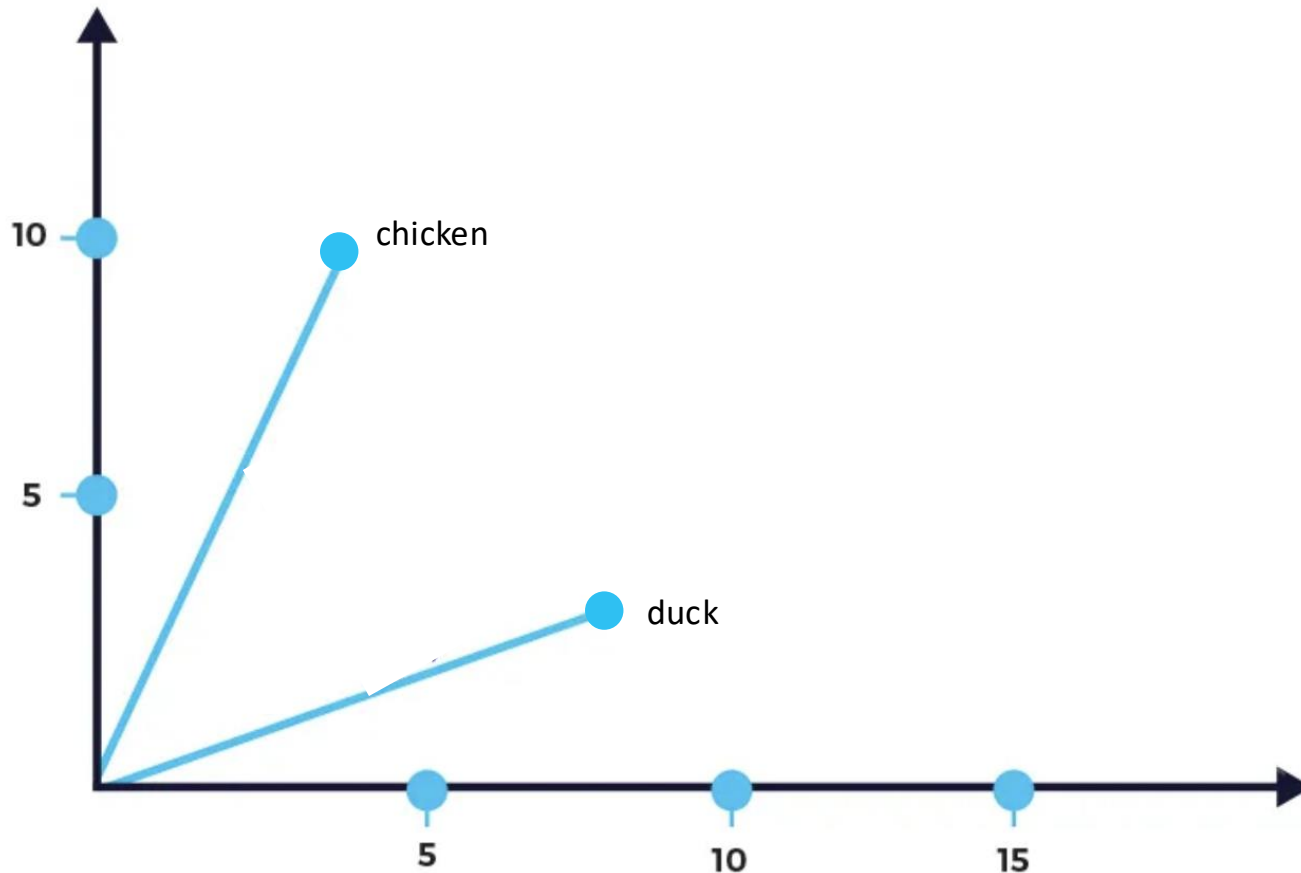


[-41.2, ..., 689.0]

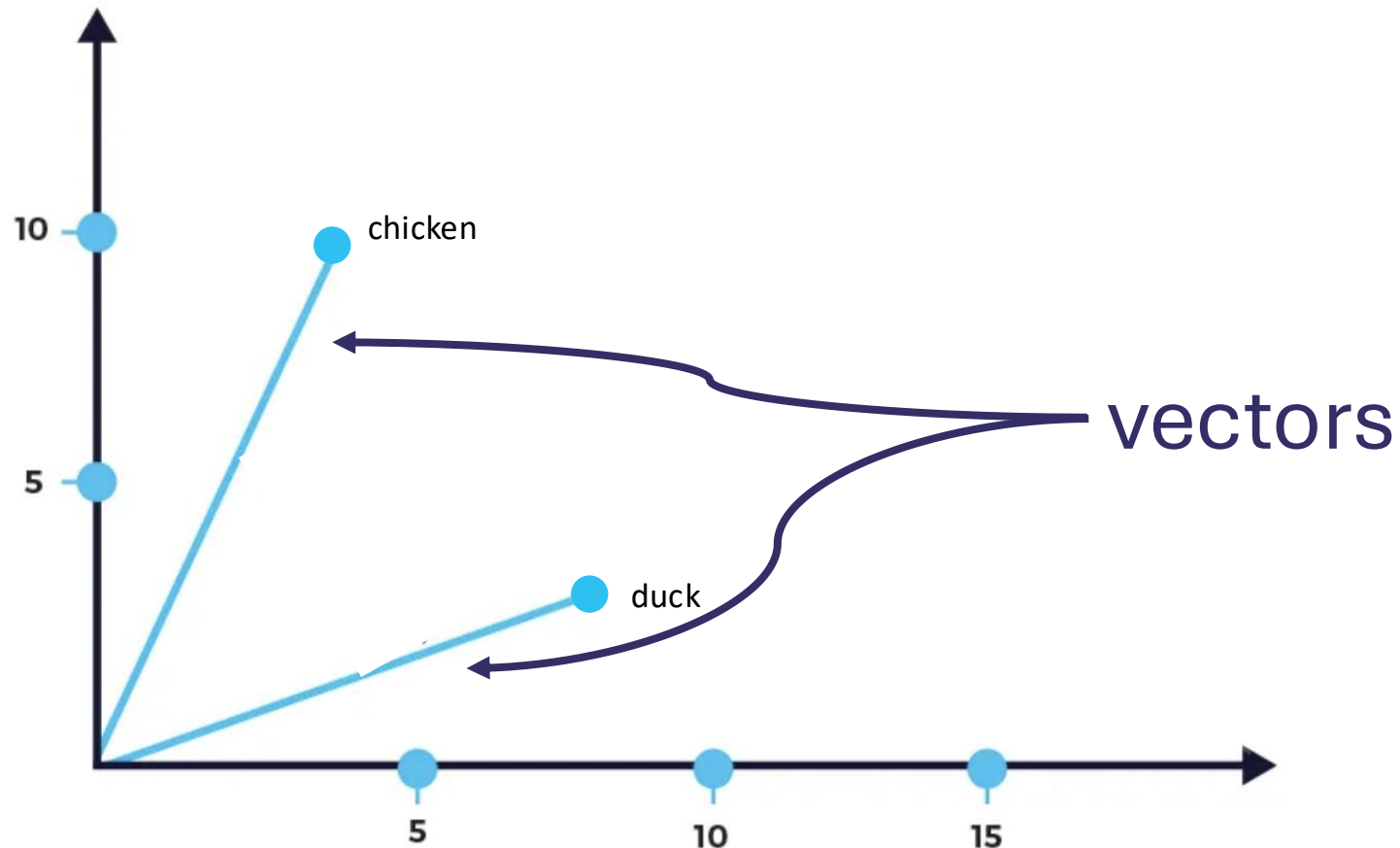
Mathematical Comparisons: Cosine Similarity



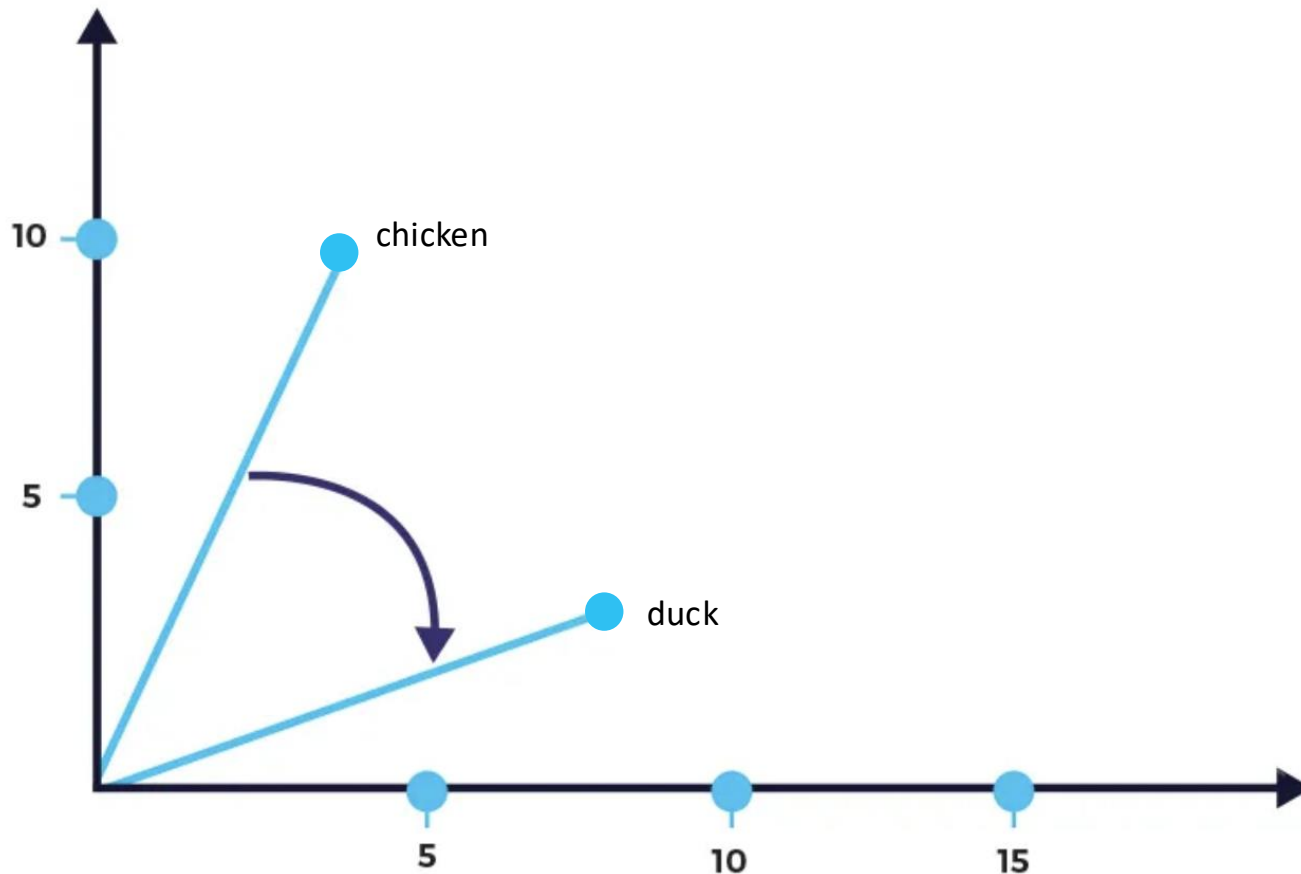
Mathematical Comparisons: Cosine Similarity



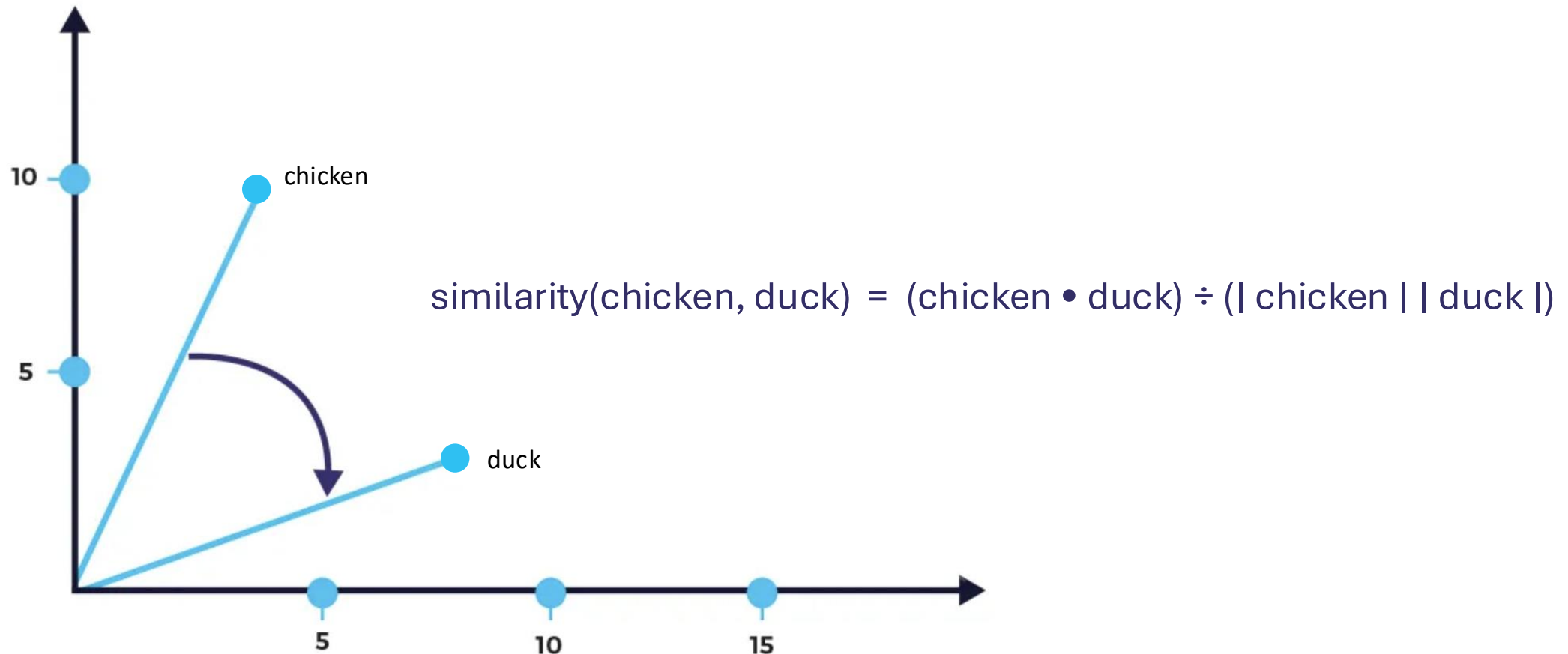
Mathematical Comparisons: Cosine Similarity



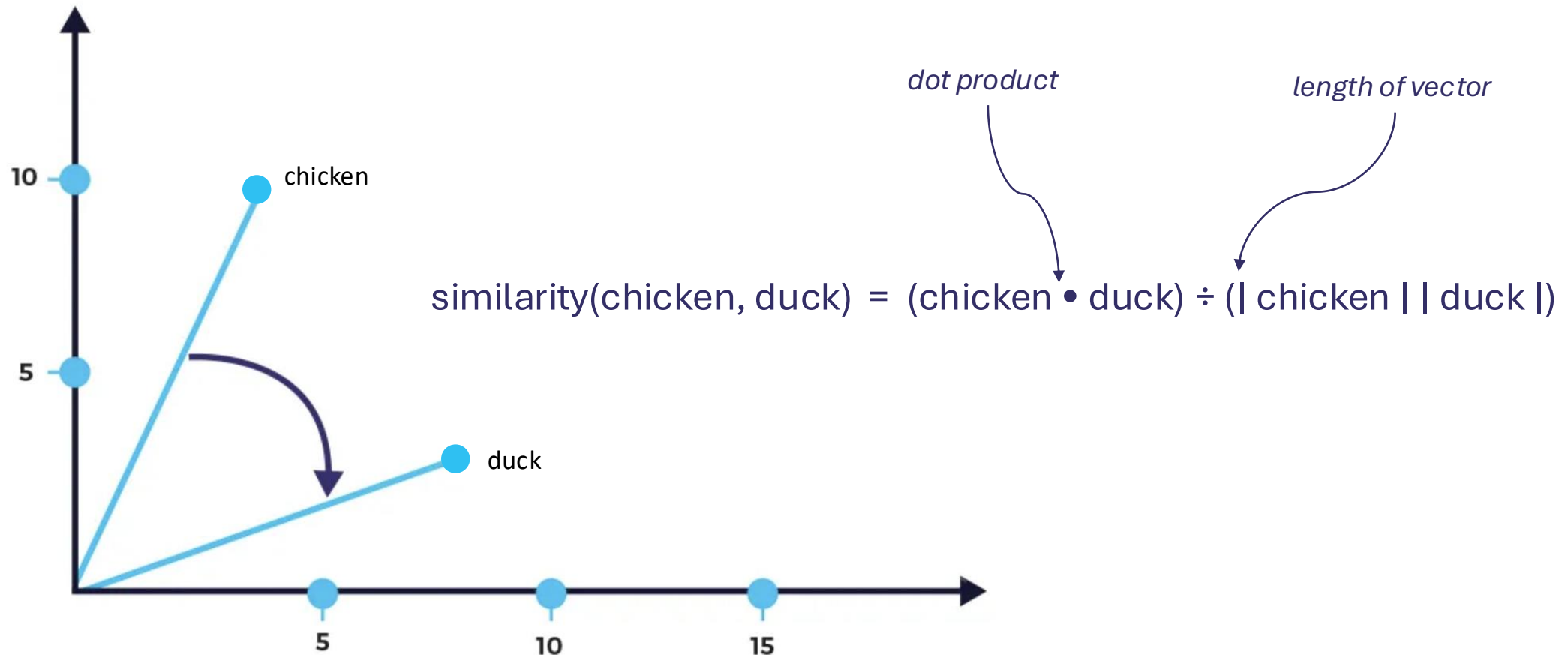
Mathematical Comparisons: Cosine Similarity



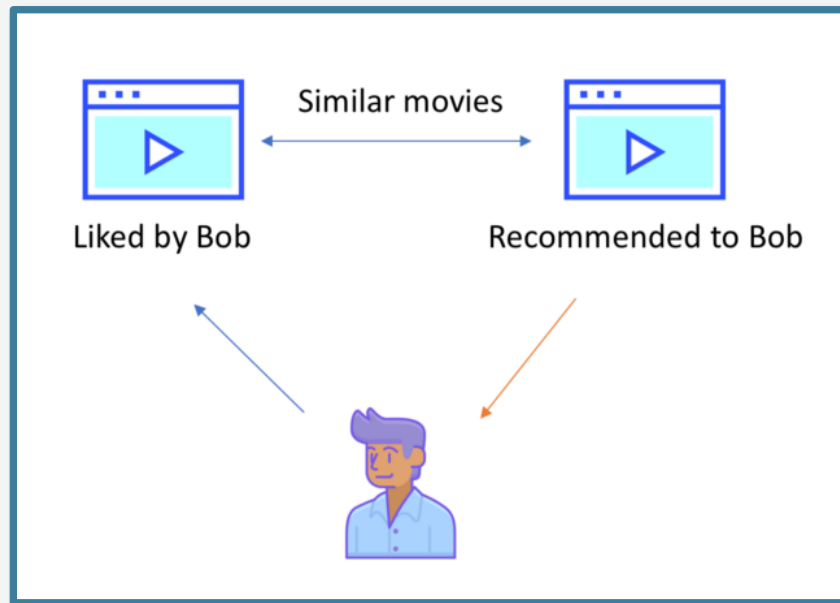
Mathematical Comparisons: Cosine Similarity



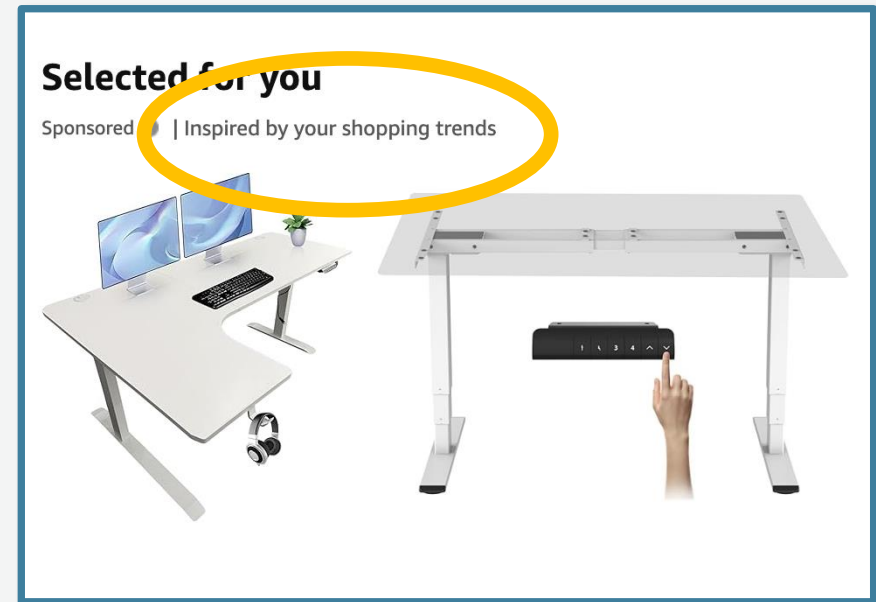
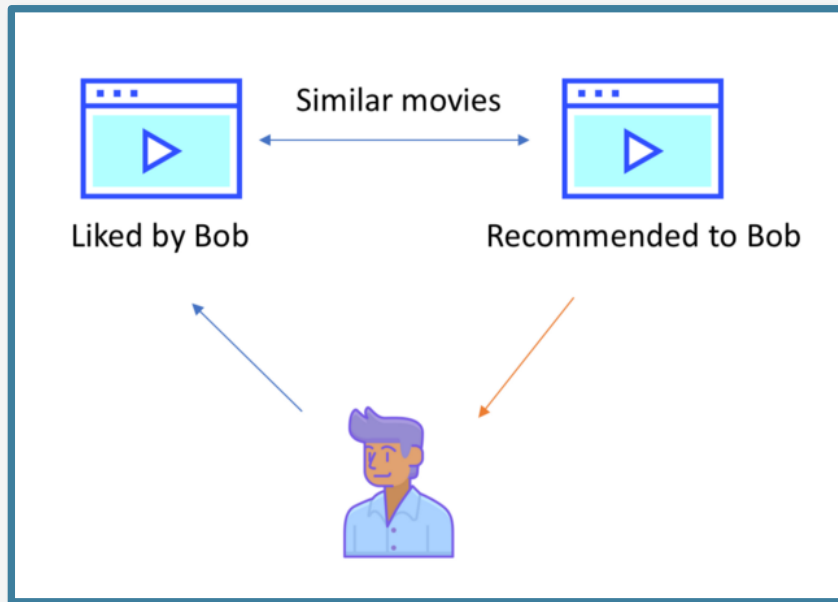
Mathematical Comparisons: Cosine Similarity



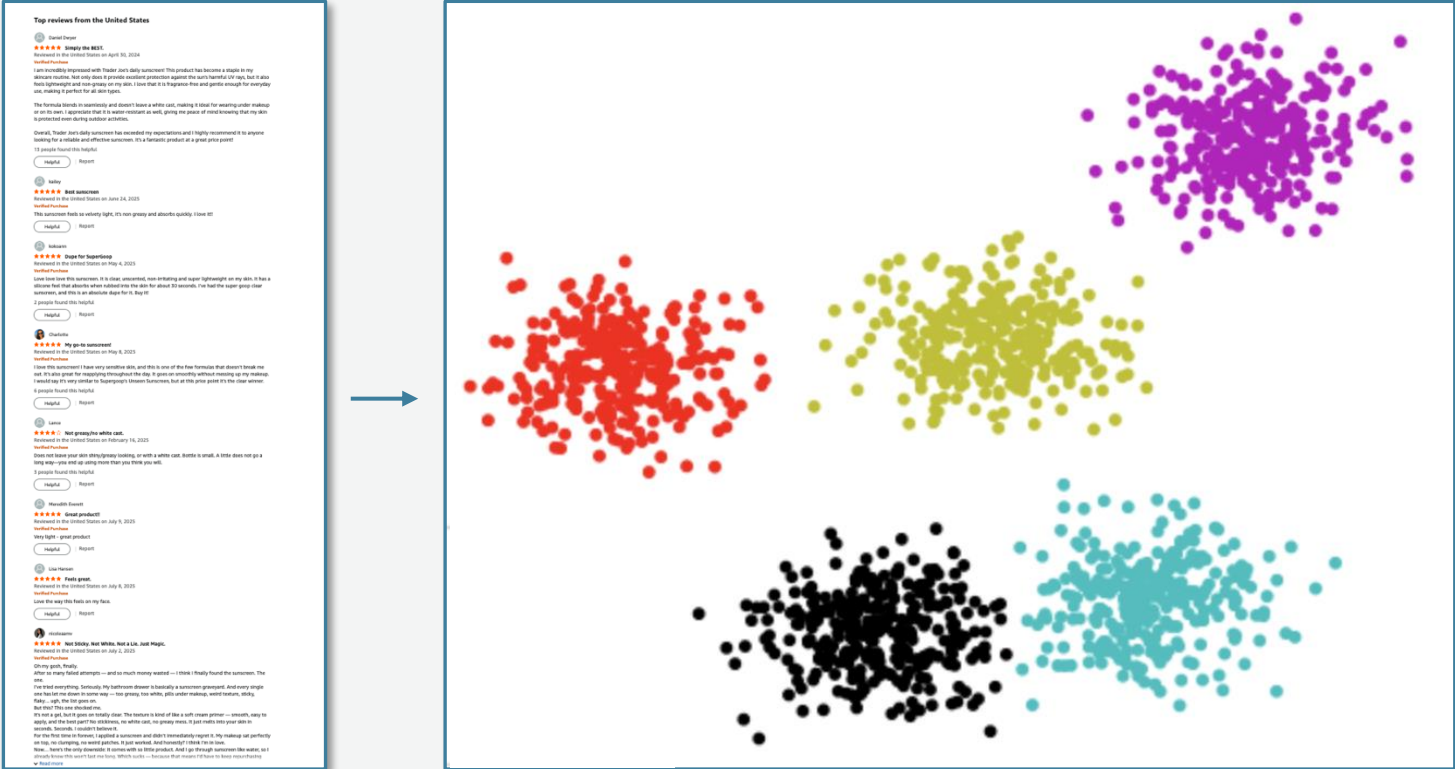
Content Recommendation



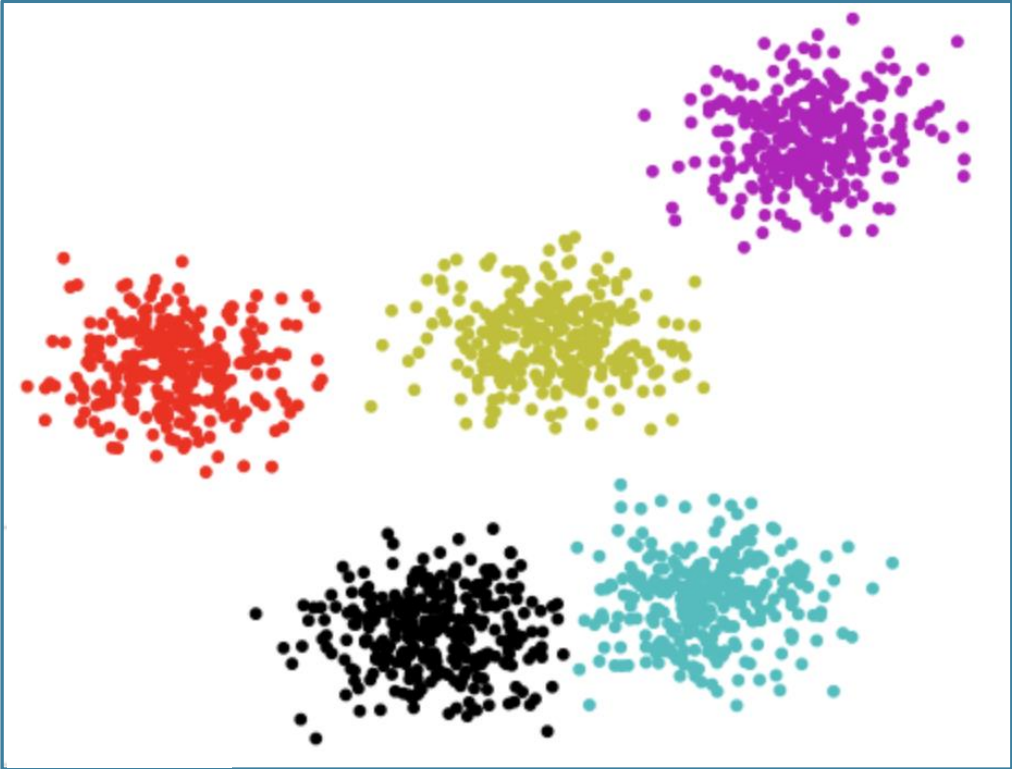
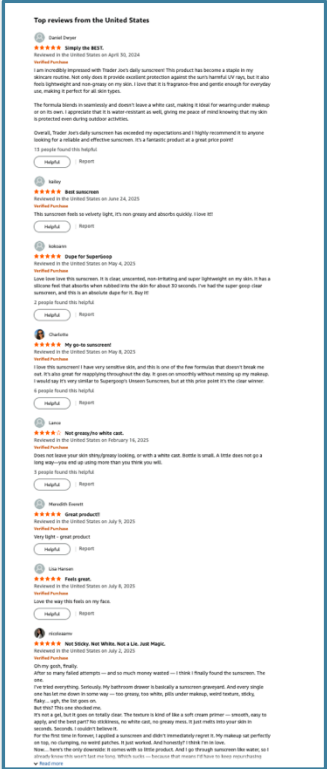
Content Recommendation



Clustering Comments and Reviews



Clustering Comments and Reviews



Select to learn more

- ✓ Sunscreen effectiveness
- ✓ Skin texture
- ✓ Lightweight
- ✓ Oiliness
- ✓ Skin protection
- ✓ Whiteness
- ✓ Smoothness

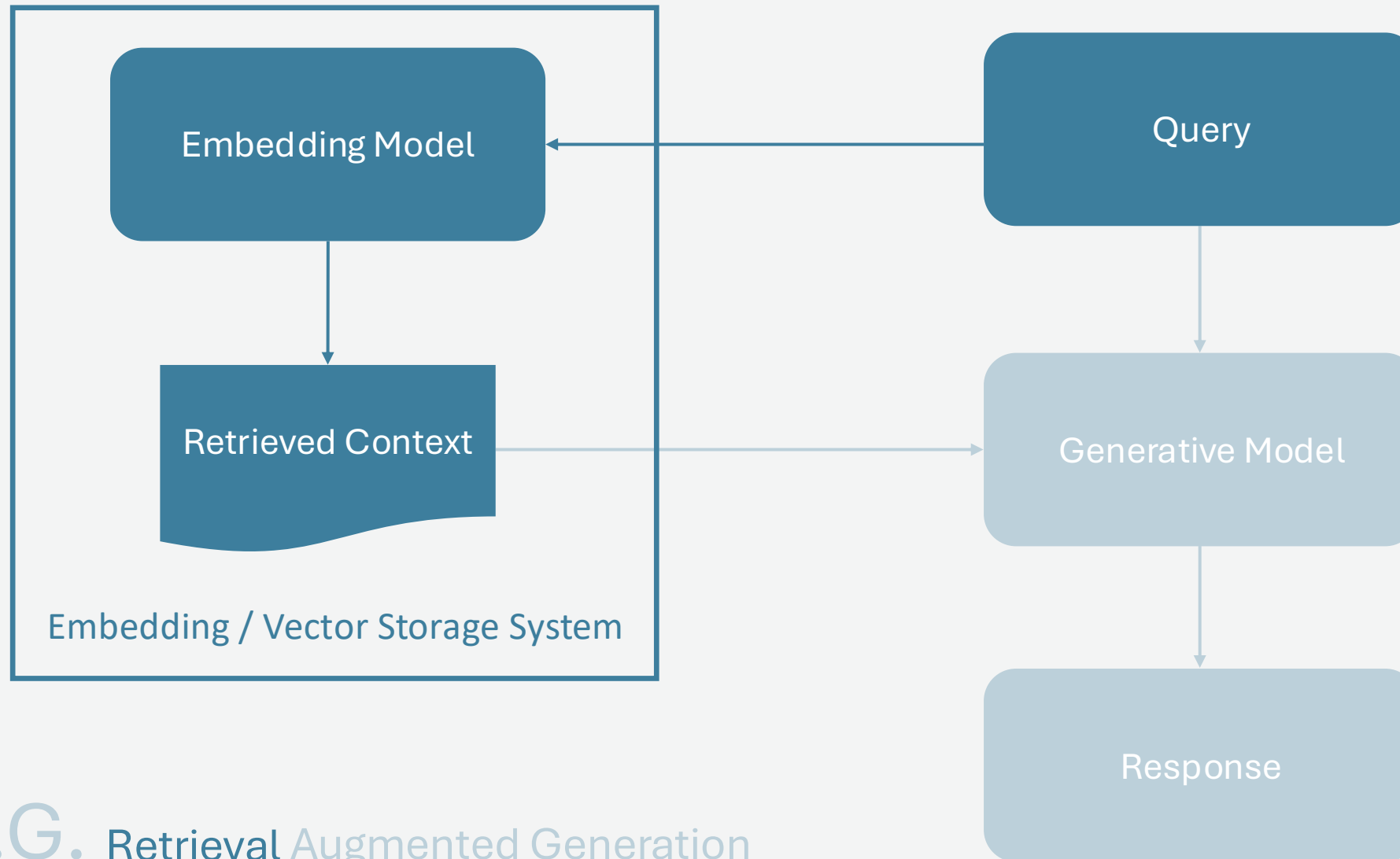
- ✓ Sunscreen effectiveness | ✓ Skin texture |
- ✓ Lightweight | ✓ Oiliness | ✓ Skin protection |
- ✓ Whiteness | ✓ Smoothness

R. Retrieval

A. Augmented

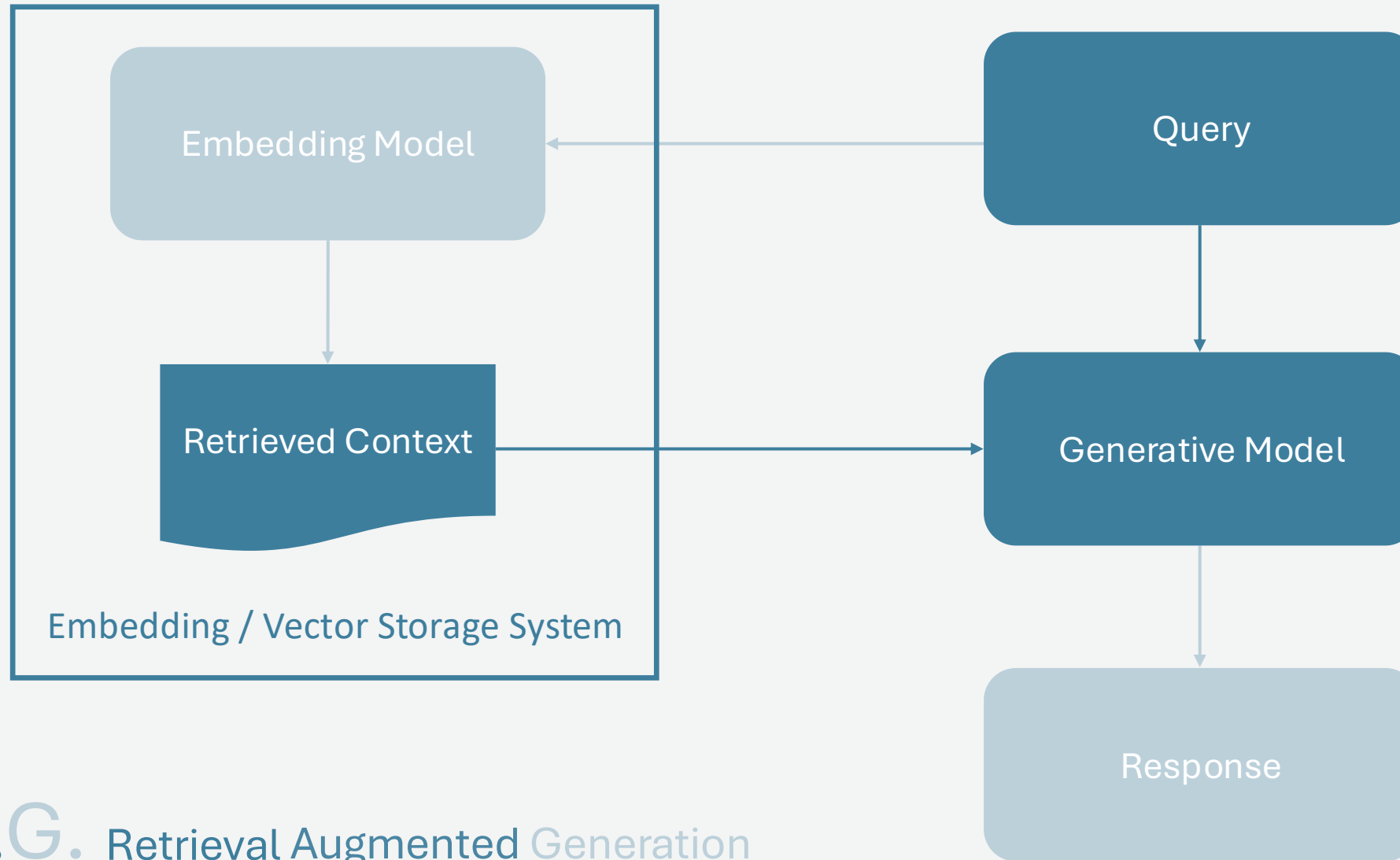
G. Generation

Other Use Cases for Embeddings



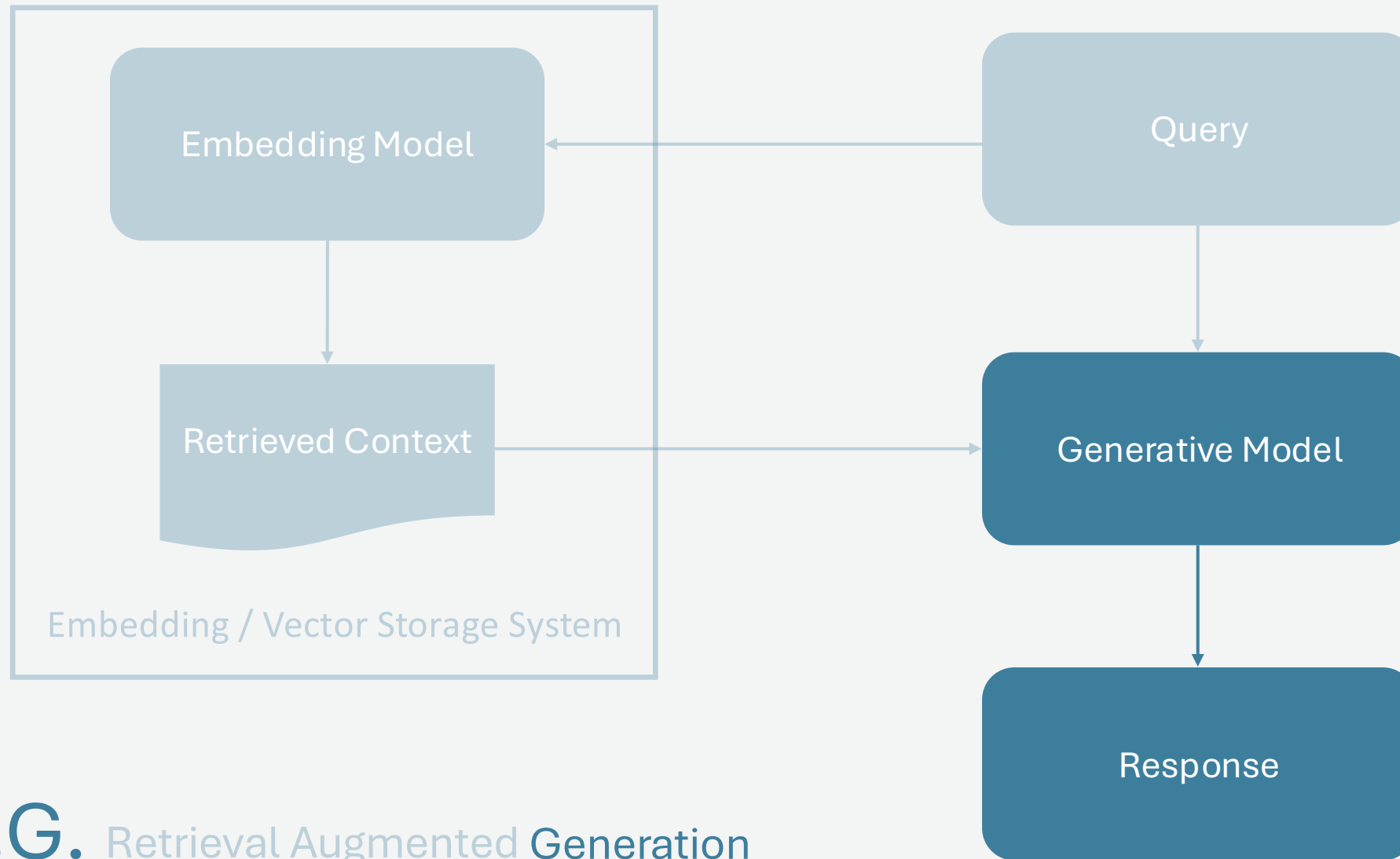
R.A.G. Retrieval Augmented Generation

Other Use Cases for Embeddings



R.A.G. Retrieval Augmented Generation

Other Use Cases for Embeddings



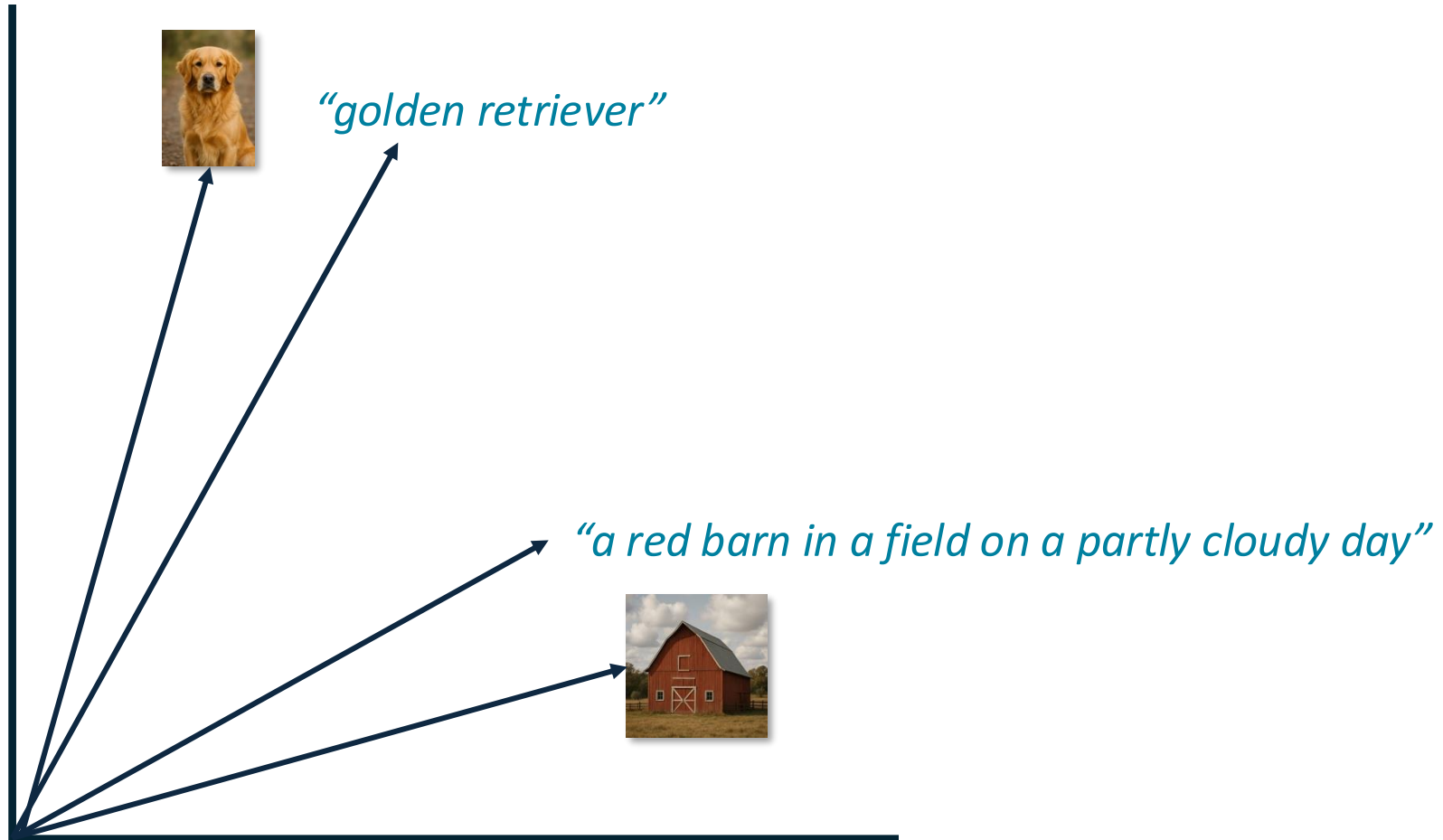
R.A.G. Retrieval Augmented Generation

Multi-Modal Embeddings

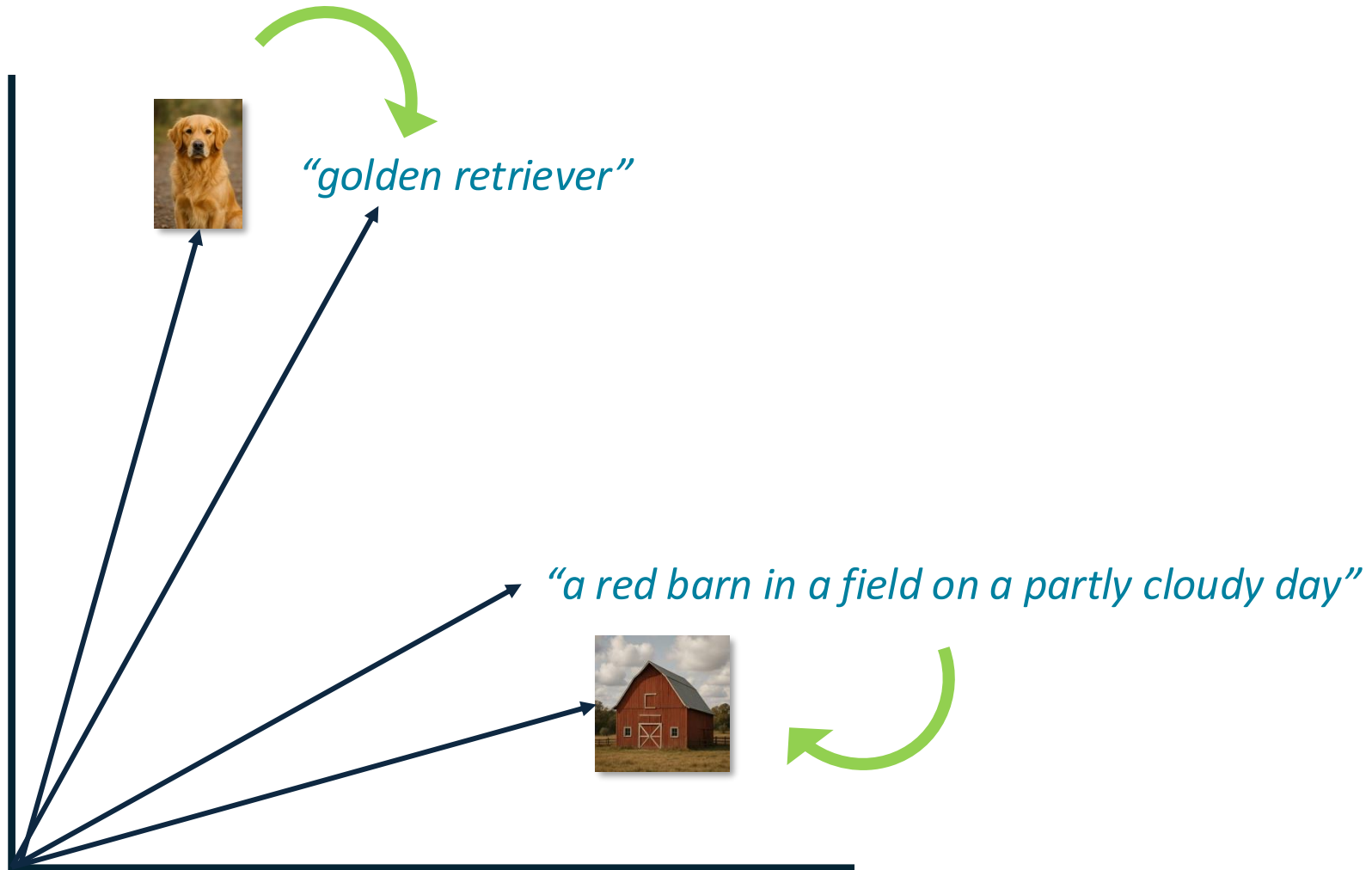


$[-0.63, \dots, 3.49]$

Multi-Modal Embeddings



Multi-Modal Embeddings



Multi-Modal Embeddings

“a golden retriever in front of a red barn”



Audience Poll Question

What cannot be embedded? (multiple choice)

- Sentences
- Images
- Sounds
- Documents
- Nothing, you can embed every medium

- **Word Embeddings** allow us to do math with human language so that we can build powerful LLMs
- Because word embeddings capture the **nuance of human language**, they allow us to create brilliant (and problematic) generative models
- The concept of embeddings can also be used for applications other than training models

Audience Poll Question

Do you think word embeddings are cool? (single choice)

- Yes
- No

Thank You! Questions?

